

1. Git Installation

(Ensure : You have a stable network connection before proceeding with the installation step.)

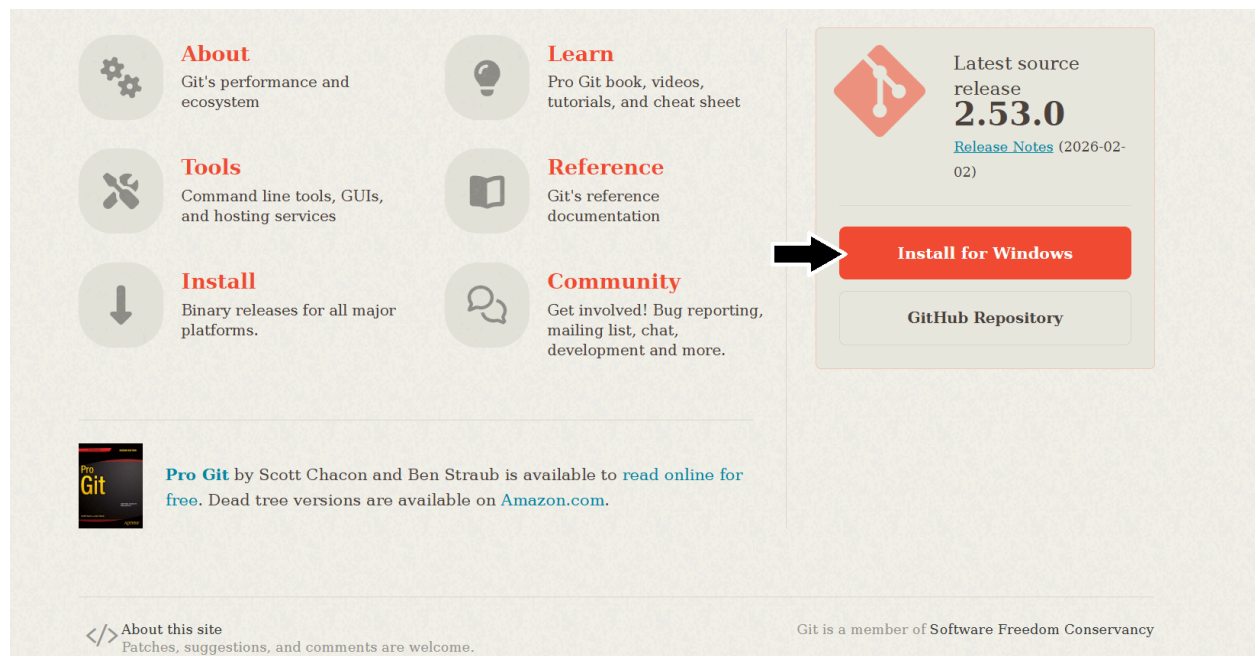
(Note : Below steps is for windows)

PART 1: Installing Git on Windows

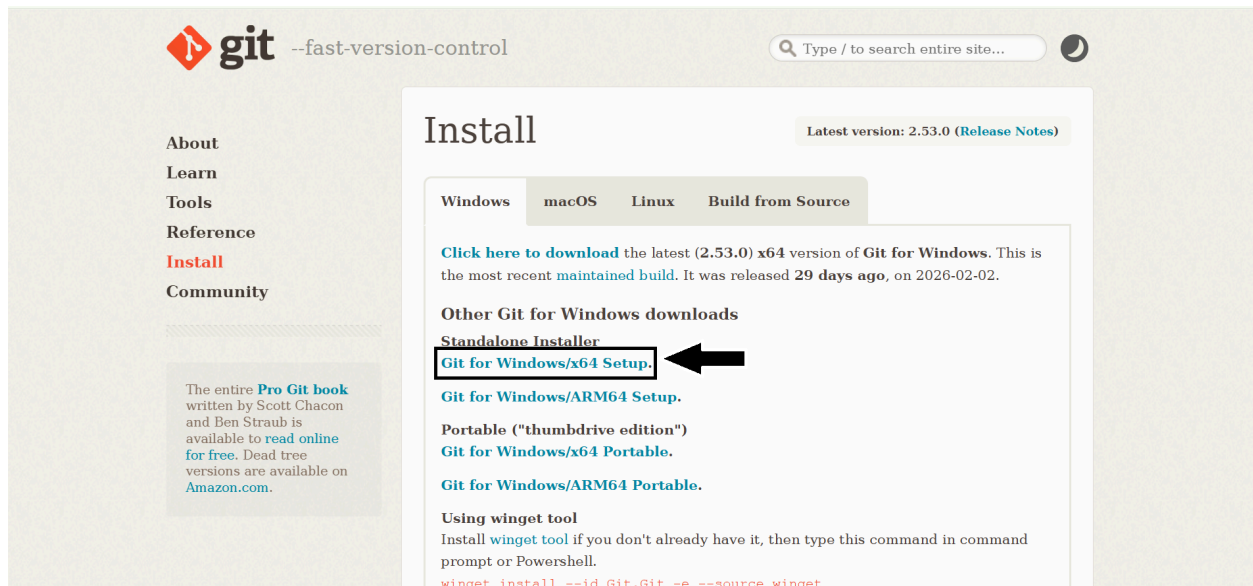
Step 1: Download Git

1. Open your browser.
2. Go to the official website:
👉 <https://git-scm.com>
3. Click on **Download for Windows**

📸 Screenshot 1: Git website download page. Click the install for windows option. (marked with arrow)



Select the option marked with an arrow.



The screenshot shows the Git website's 'Install' page. The header includes the Git logo, the tagline '--fast-version-control', a search bar, and a dark mode toggle. A left sidebar contains navigation links: 'About', 'Learn', 'Tools', 'Reference', 'Install' (highlighted in red), and 'Community'. Below these links is a text box mentioning the 'Pro Git book'. The main content area is titled 'Install' and shows the 'Latest version: 2.53.0 (Release Notes)'. It features tabs for 'Windows', 'macOS', 'Linux', and 'Build from Source', with 'Windows' selected. The text instructs users to 'Click here to download' the latest (2.53.0) x64 version of Git for Windows, noting it was released 29 days ago. Under 'Other Git for Windows downloads', there is a section for 'Standalone Installer' with a link 'Git for Windows/x64 Setup.' highlighted by a black box and a black arrow pointing to it. Other links include 'Git for Windows/ARM64 Setup.', 'Portable ("thumbdrive edition")', 'Git for Windows/x64 Portable.', and 'Git for Windows/ARM64 Portable.'. A section titled 'Using winget tool' provides instructions to install the winget tool and a command to install Git: `winget install --id Git.Git -e --source winget`.

git --fast-version-control

Type / to search entire site...

About
Learn
Tools
Reference
Install
Community

The entire [Pro Git book](#) written by Scott Chacon and Ben Straub is available to [read online](#) for free. Dead tree versions are available on [Amazon.com](#).

Install

Latest version: 2.53.0 ([Release Notes](#))

Windows macOS Linux Build from Source

Click [here to download](#) the latest (2.53.0) x64 version of **Git for Windows**. This is the most recent [maintained build](#). It was released **29 days ago**, on 2026-02-02.

Other Git for Windows downloads

Standalone Installer

[Git for Windows/x64 Setup.](#)

[Git for Windows/ARM64 Setup.](#)

Portable ("thumbdrive edition")

[Git for Windows/x64 Portable.](#)

[Git for Windows/ARM64 Portable.](#)

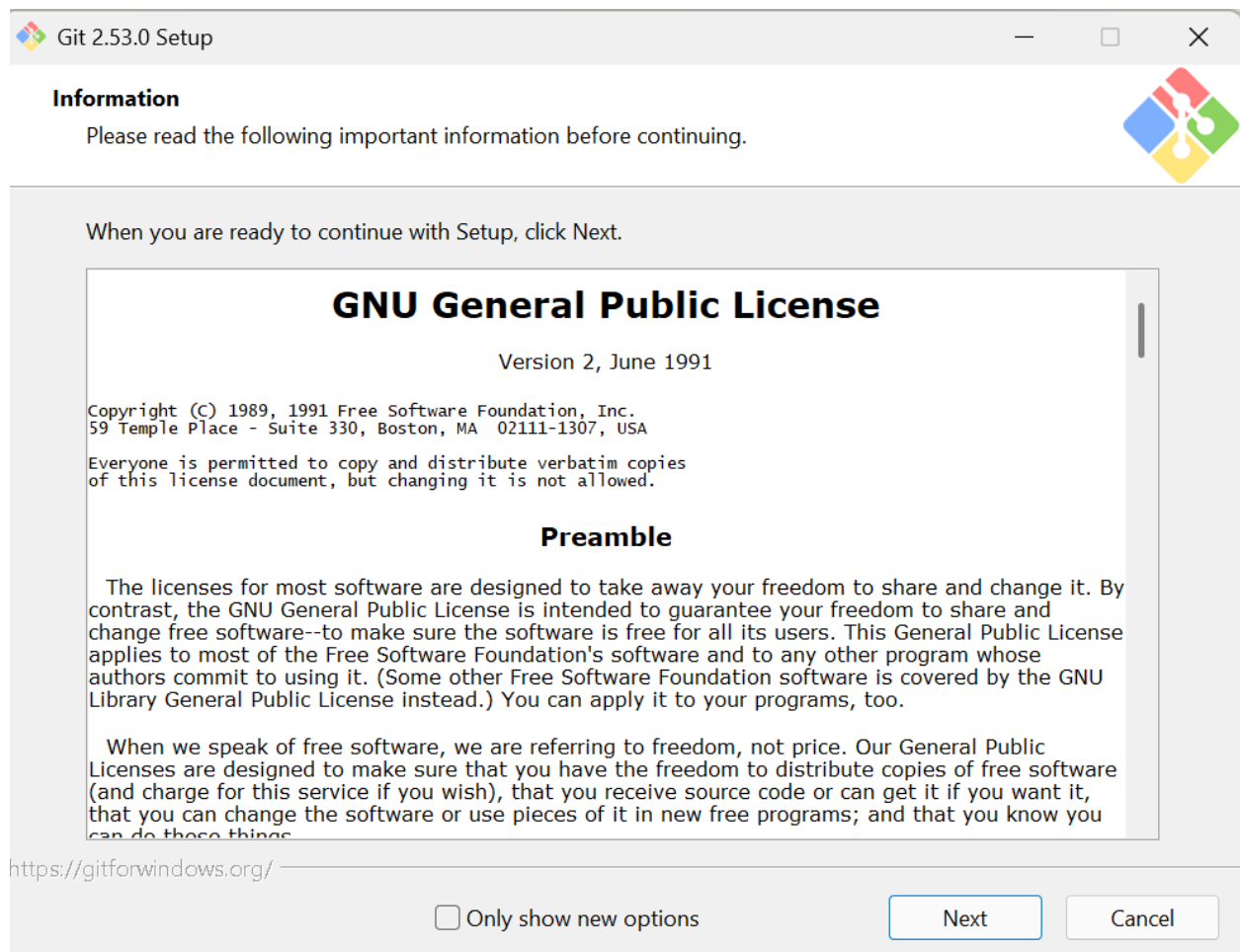
Using winget tool

Install [winget tool](#) if you don't already have it, then type this command in command prompt or Powershell.

```
winget install --id Git.Git -e --source winget
```

Step 2: Run the Installer

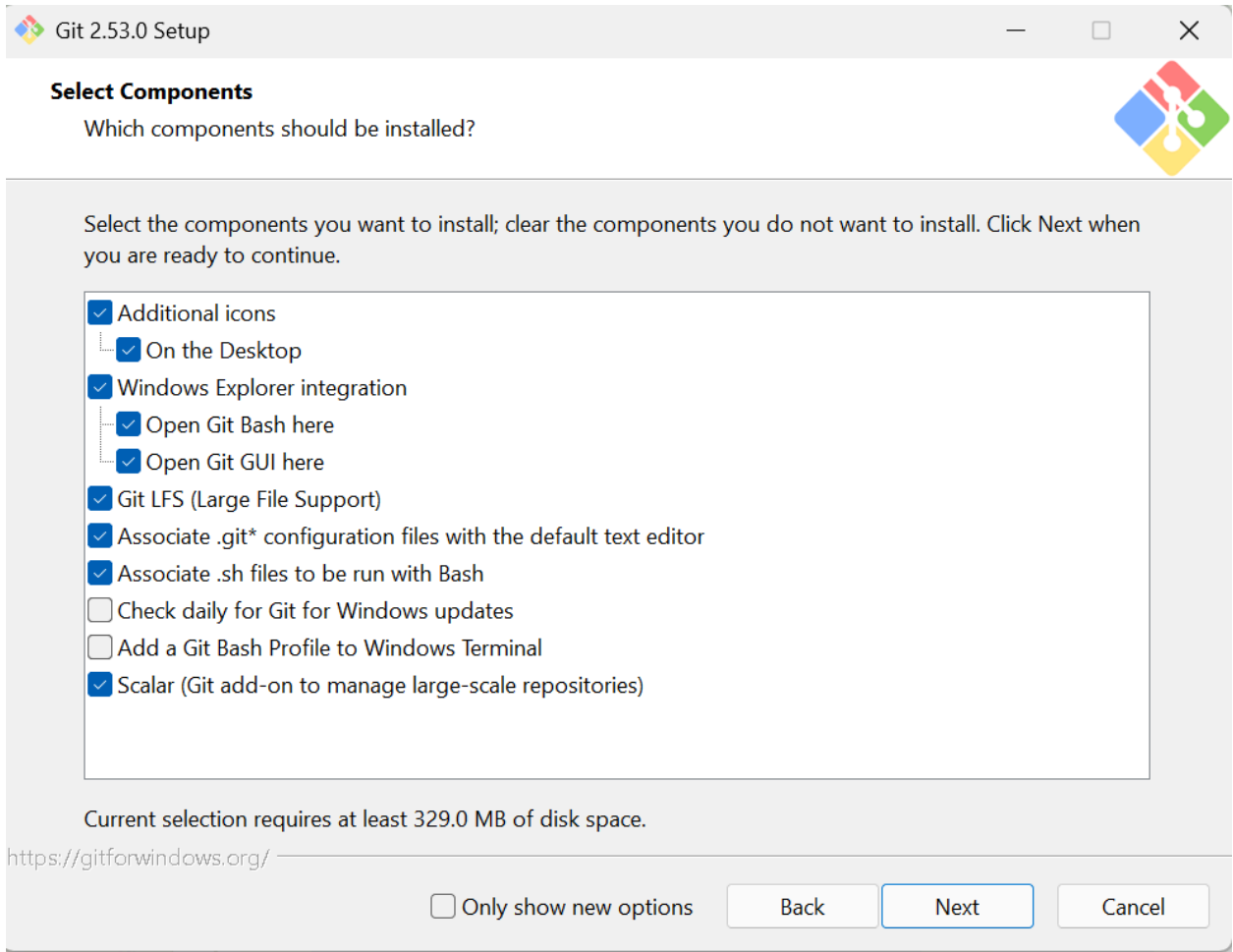
1. Double-click the downloaded file.
2. Click **Next** on the welcome screen.



Step 3: Select Components

Keep everything as **default**.

Just click **Next**.



Git 2.53.0 Setup

Select Components

Which components should be installed?

Select the components you want to install; clear the components you do not want to install. Click Next when you are ready to continue.

- ☒ Additional icons
 - ☒ On the Desktop
- ☒ Windows Explorer integration
 - ☒ Open Git Bash here
 - ☒ Open Git GUI here
- ☒ Git LFS (Large File Support)
- ☒ Associate .git* configuration files with the default text editor
- ☒ Associate .sh files to be run with Bash
- ☐ Check daily for Git for Windows updates
- ☐ Add a Git Bash Profile to Windows Terminal
- ☒ Scalar (Git add-on to manage large-scale repositories)

Current selection requires at least 329.0 MB of disk space.

<https://gitforwindows.org/>

☐ Only show new options

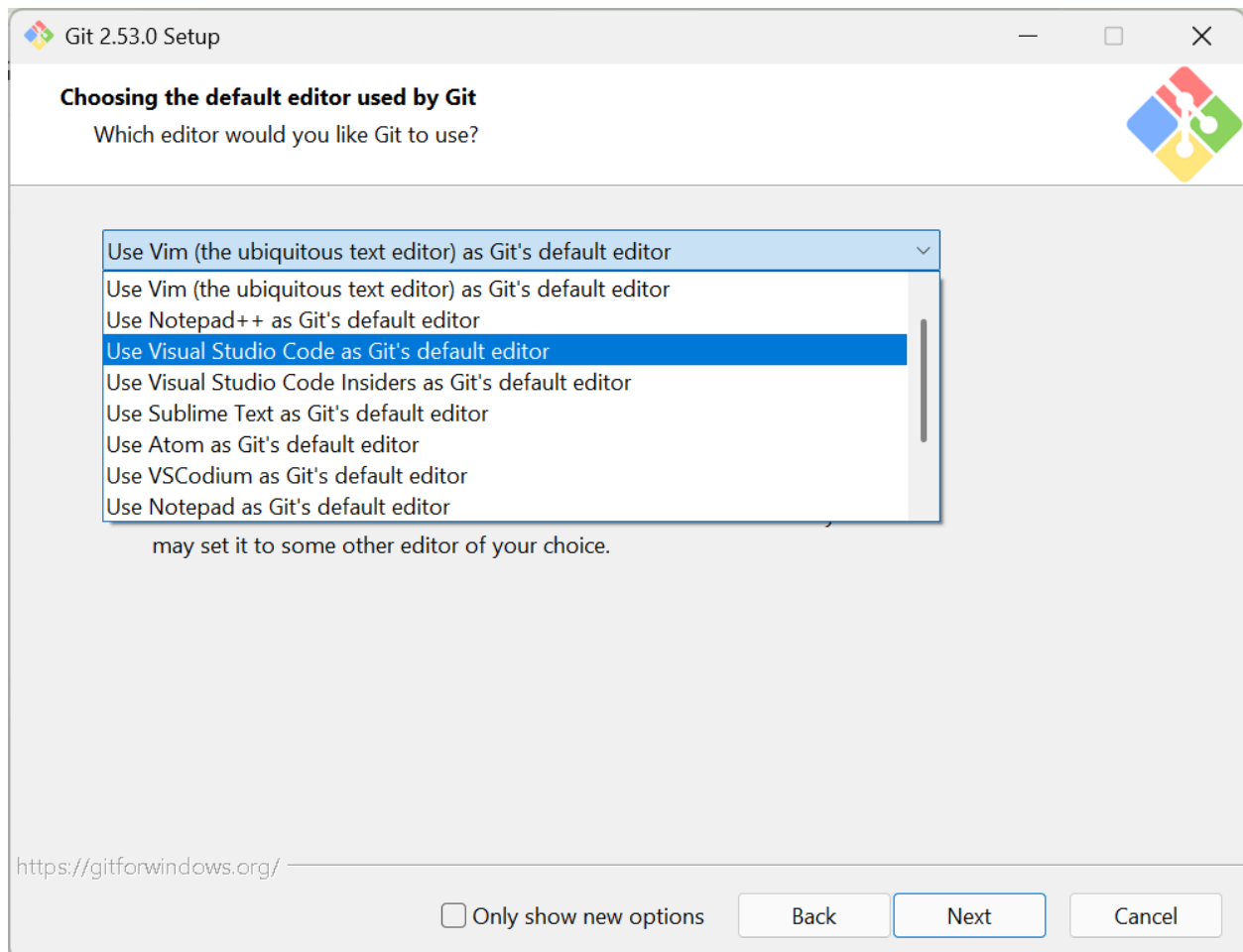
Back Next Cancel

Step 4: Choose Default Editor

When asked for default editor:

👉 Select **Use Visual Studio Code as Git's default editor** (If VS Code is installed)

Otherwise keep default (Vim).



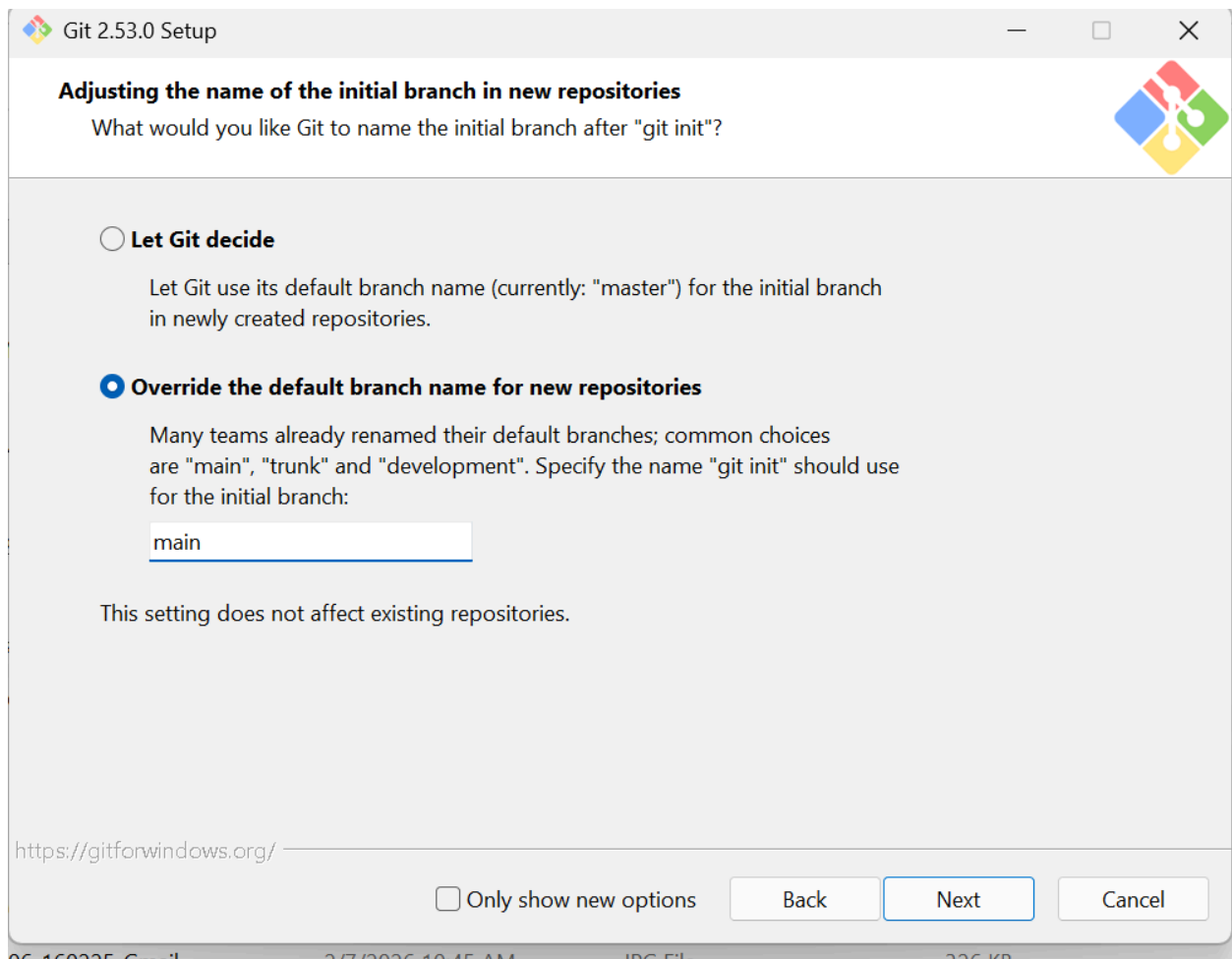
Step 5 : Choose Initial Branch Name

👉 Select:

Override the default branch name for new repositories
and keep it as:

main

Then click **Next**



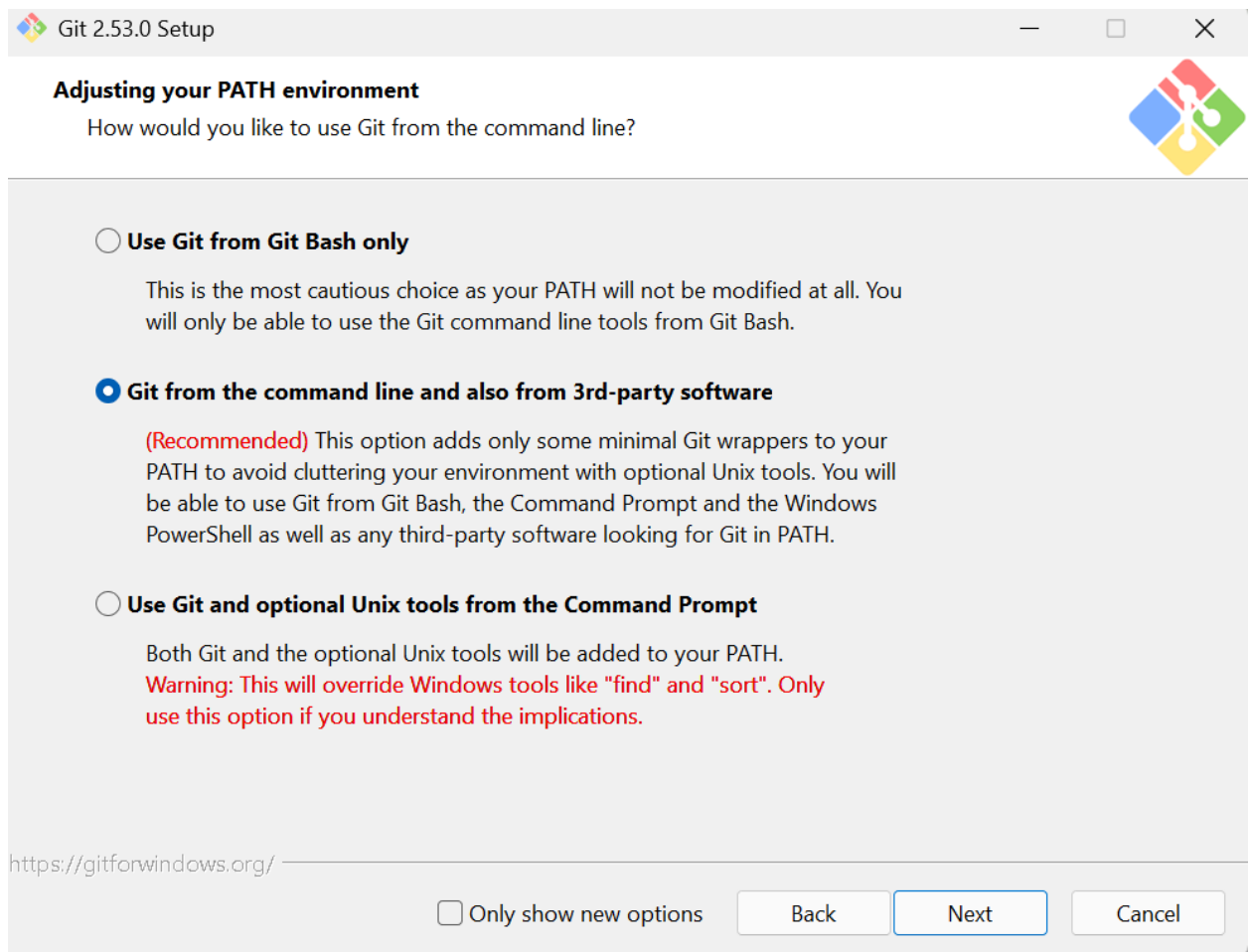
Step 6: Adjust PATH Environment

⚠ IMPORTANT STEP

Select:

👉 **Git from the command line and also from 3rd-party software**

Then click **Next**



The image shows a screenshot of the 'Git 2.53.0 Setup' window. The title bar reads 'Git 2.53.0 Setup'. The main heading is 'Adjusting your PATH environment'. Below this, it asks 'How would you like to use Git from the command line?'. There are three radio button options:

- ☐ **Use Git from Git Bash only**
This is the most cautious choice as your PATH will not be modified at all. You will only be able to use the Git command line tools from Git Bash.
- ☒ **Git from the command line and also from 3rd-party software**
(Recommended) This option adds only some minimal Git wrappers to your PATH to avoid cluttering your environment with optional Unix tools. You will be able to use Git from Git Bash, the Command Prompt and the Windows PowerShell as well as any third-party software looking for Git in PATH.
- ☐ **Use Git and optional Unix tools from the Command Prompt**
Both Git and the optional Unix tools will be added to your PATH.
Warning: This will override Windows tools like "find" and "sort". Only use this option if you understand the implications.

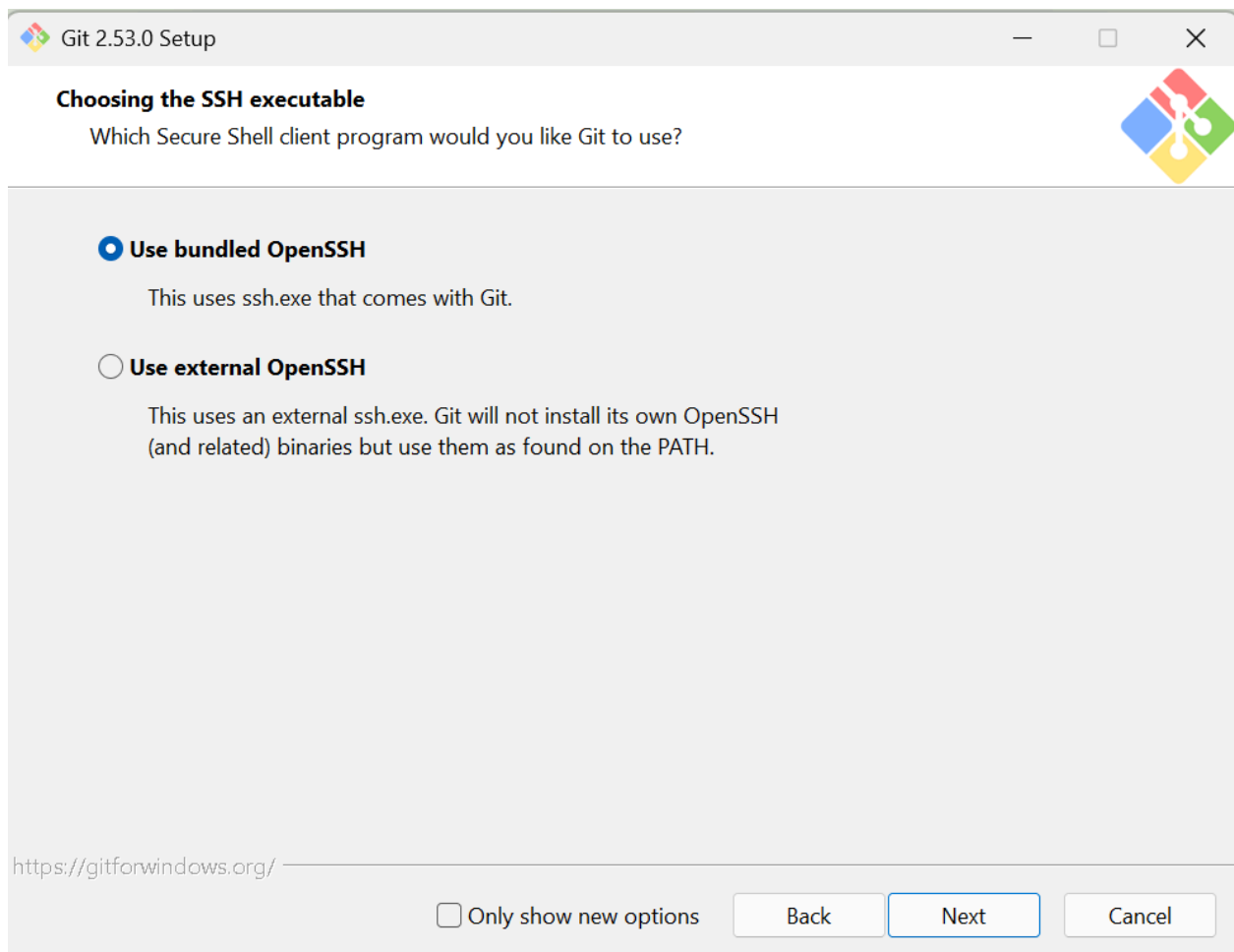
At the bottom left, there is a URL: <https://gitforwindows.org/>. At the bottom right, there are three buttons: 'Back', 'Next' (highlighted with a blue border), and 'Cancel'. There is also a checkbox labeled 'Only show new options'.

Step 7: Choose the SSH Executable

👉 Select:

Use bundled OpenSSH (Keep default)

Then click **Next**

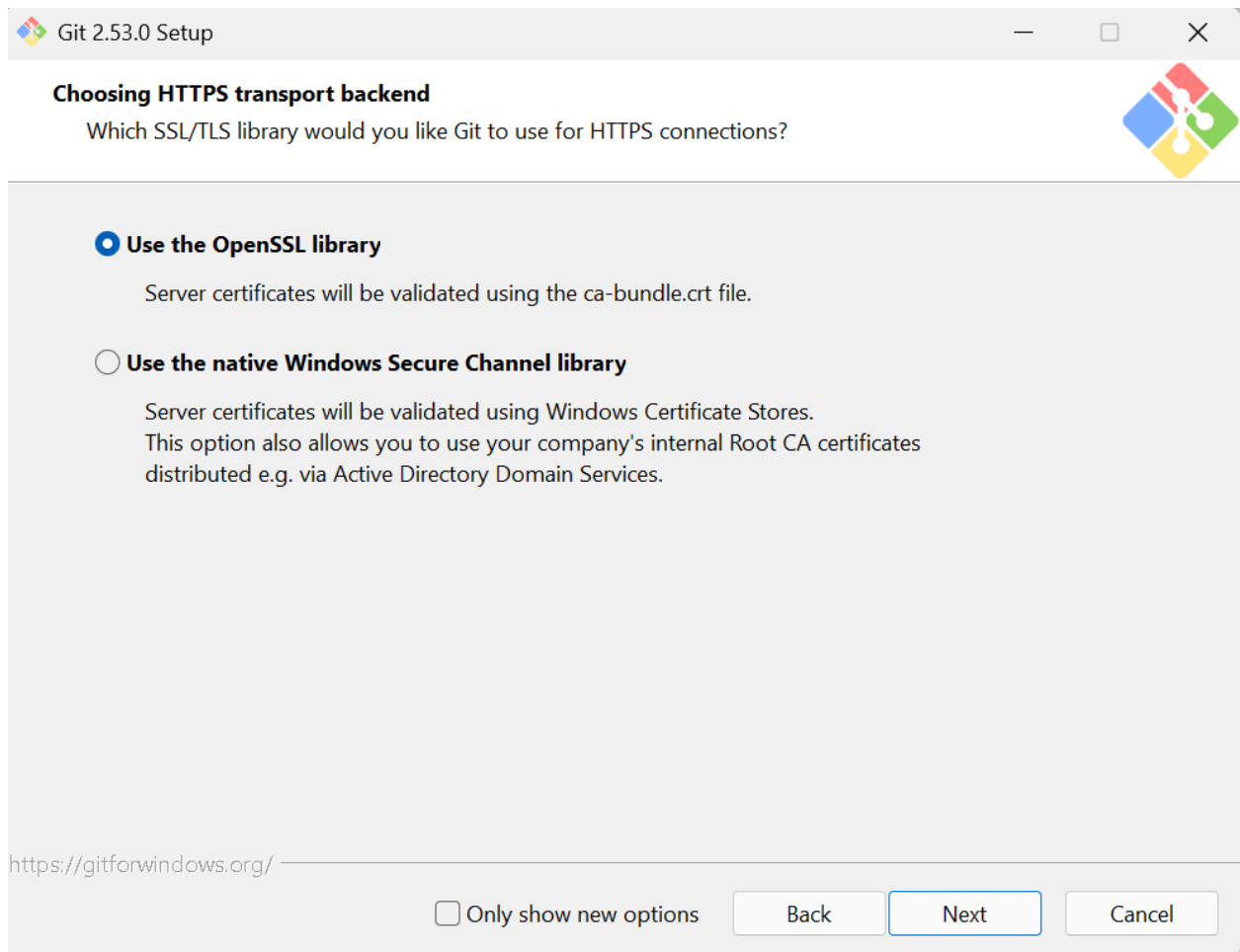


Step 8: HTTPS Transport Backend

Keep default:

👉 Use the OpenSSL library

Click **Next**



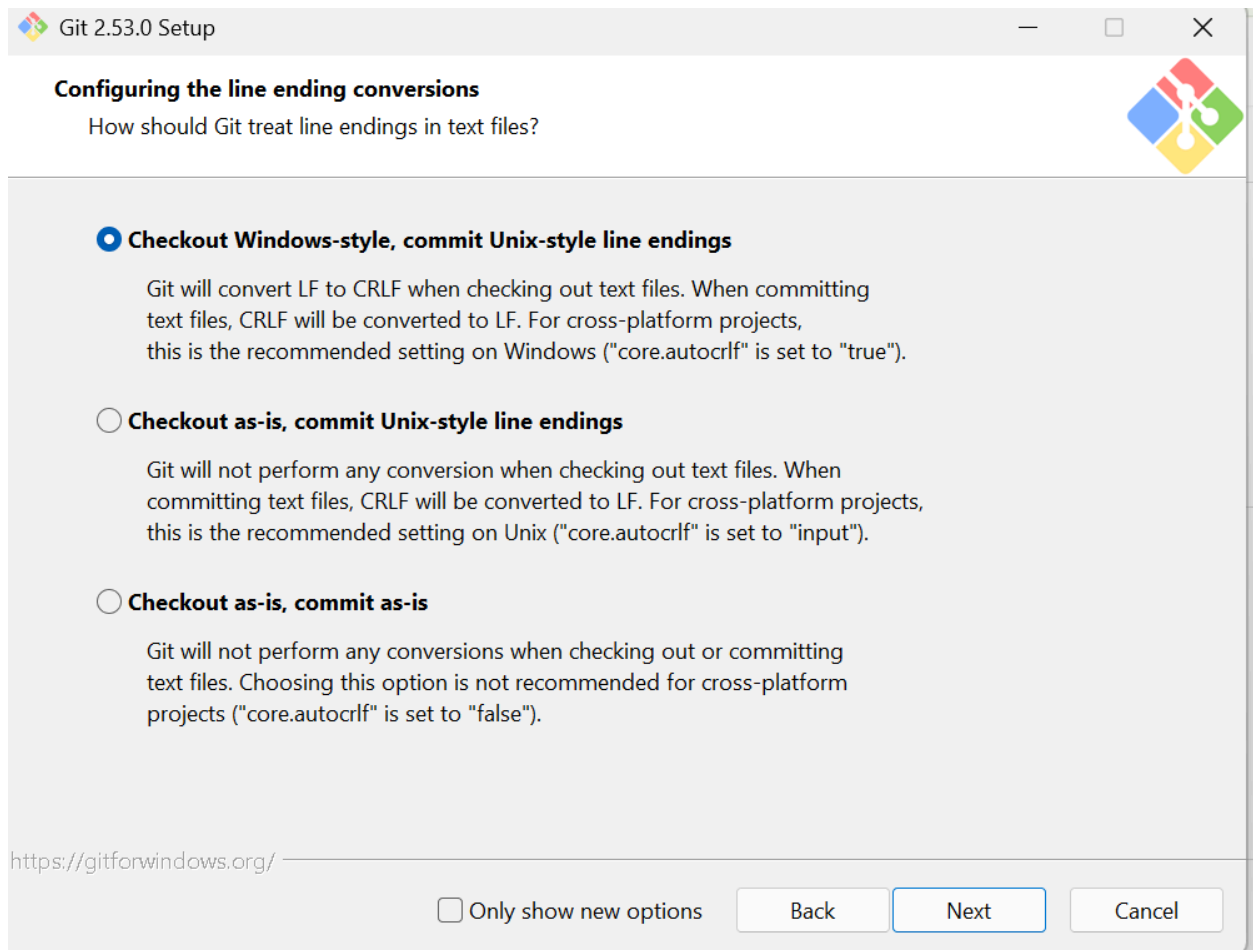
Step 9: Configuring Line Ending Conversions

Keep default:

👉 Select:

Checkout Windows-style, commit Unix-style line endings

Then click **Next**



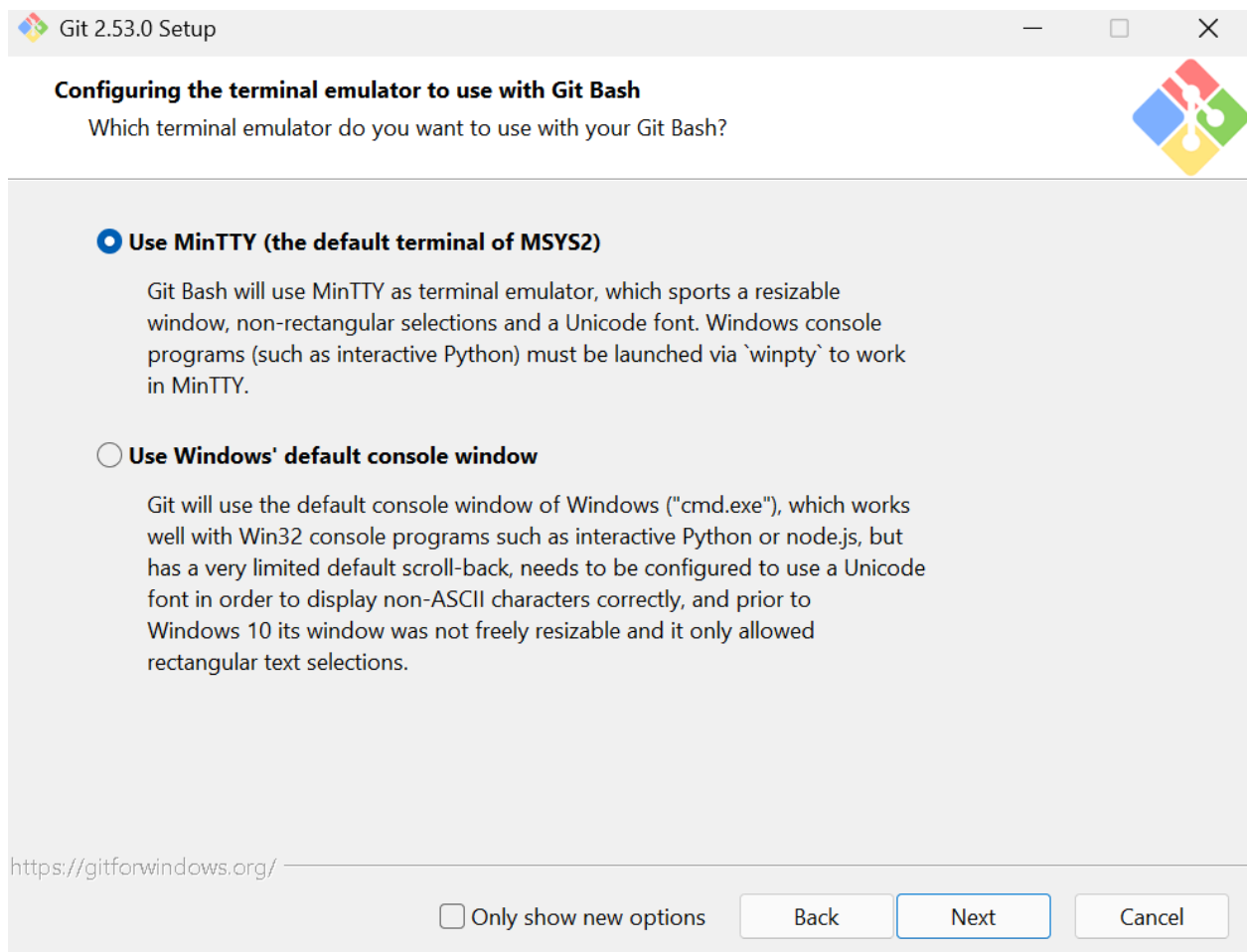
Step 10: Configuring Terminal Emulator for Git Bash

Keep default:

👉 Select:

Use MinTTY

Then click **Next**



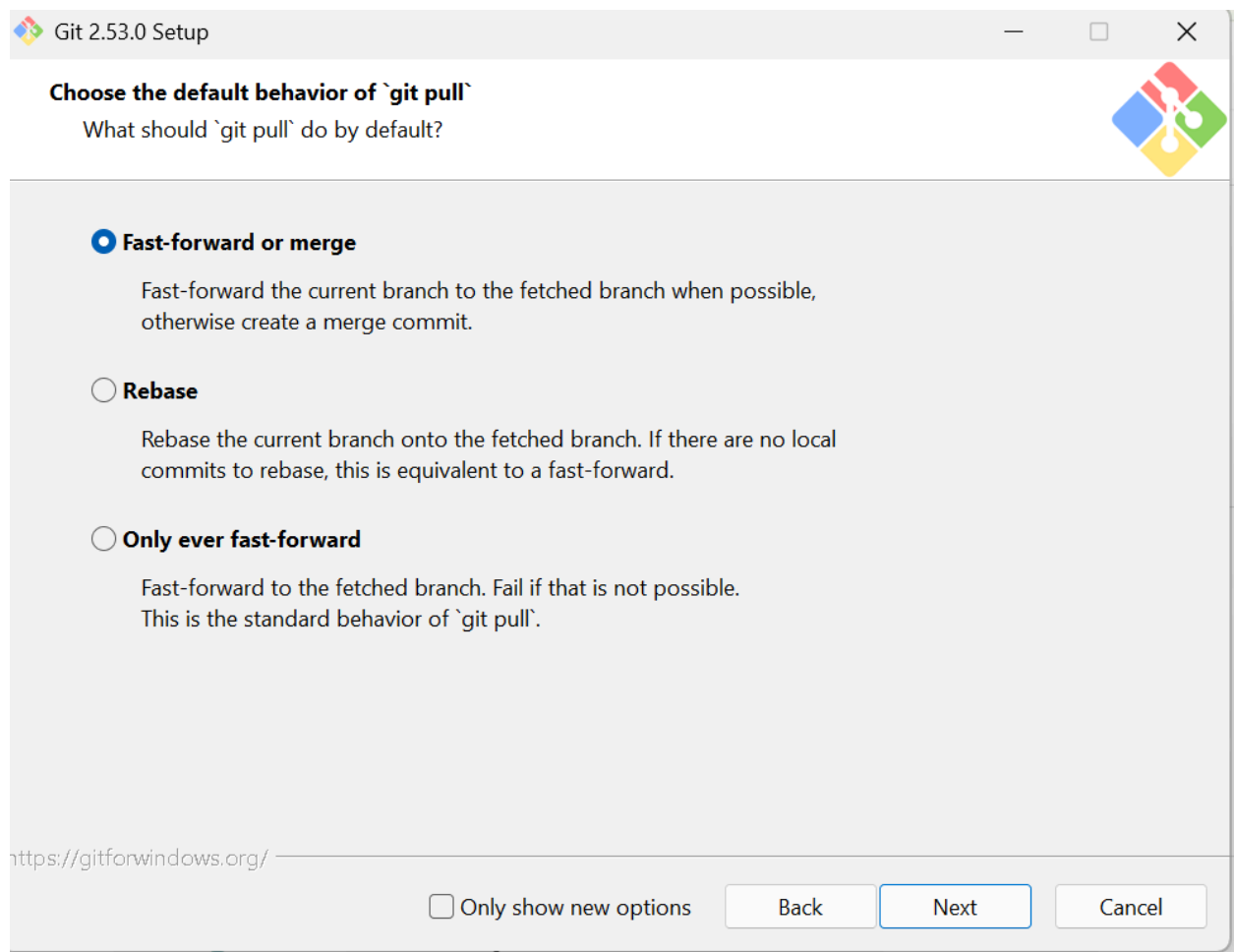
Step 11: Choose Default Behavior of git pull

Keep default:

👉 Select:

fast - forward or merge

Then click **Next**



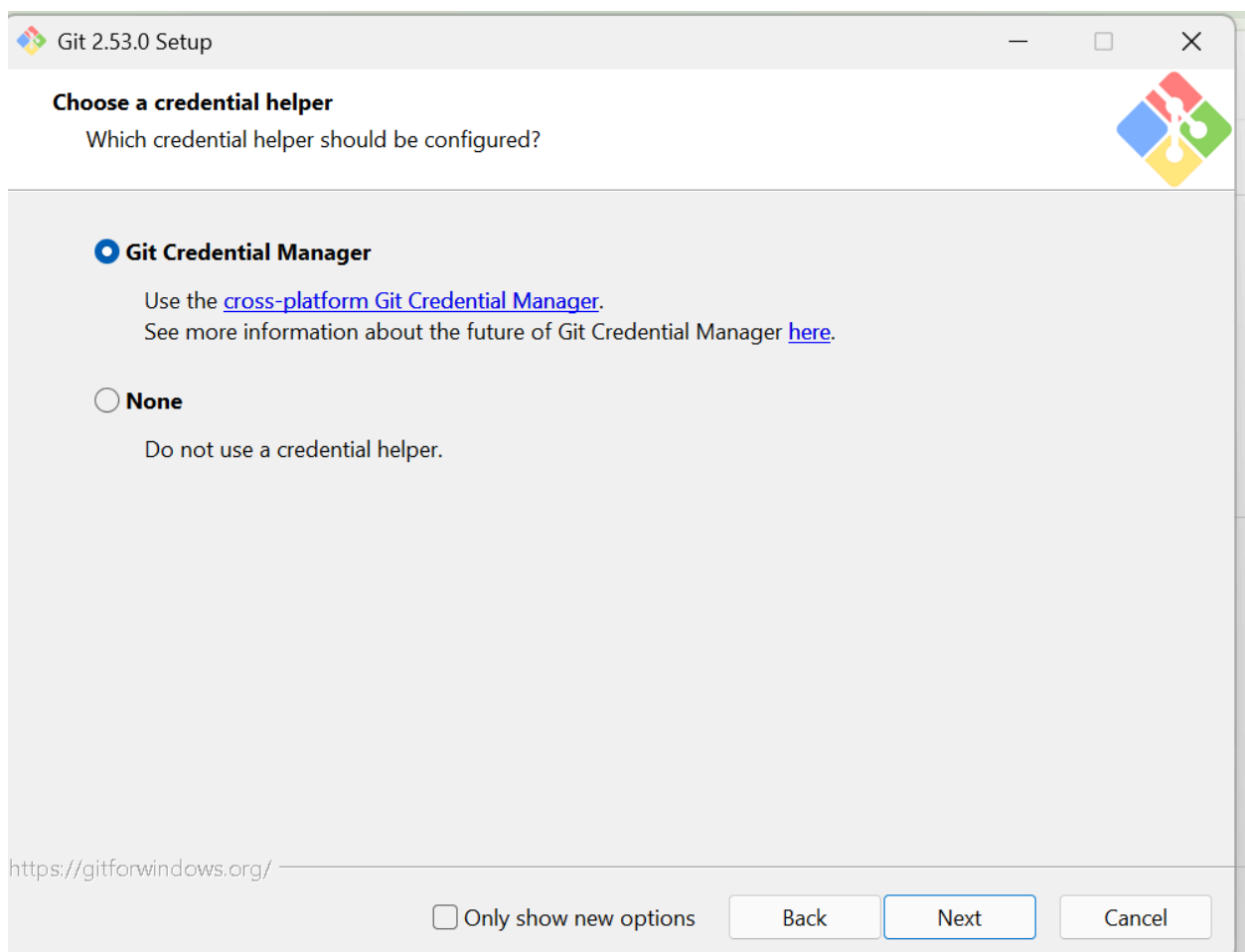
Step 12: Choose a Credential Helper

Keep default:

👉 Select:

Git Credential Manager

Then click **Next**



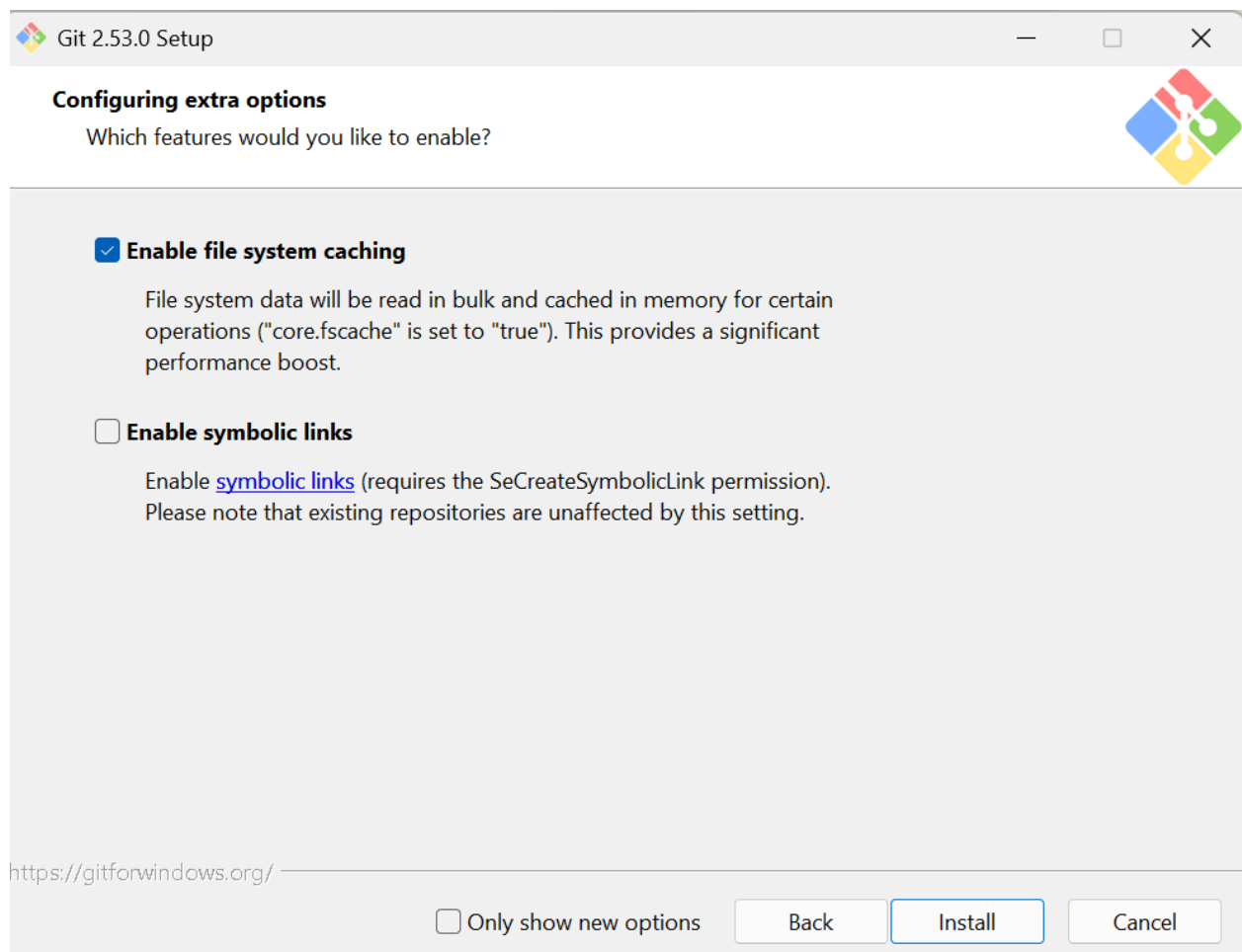
Step 13: Configuring Extra Options

Keep default:

👉 Select:

Enable file system caching

Then click **Install**

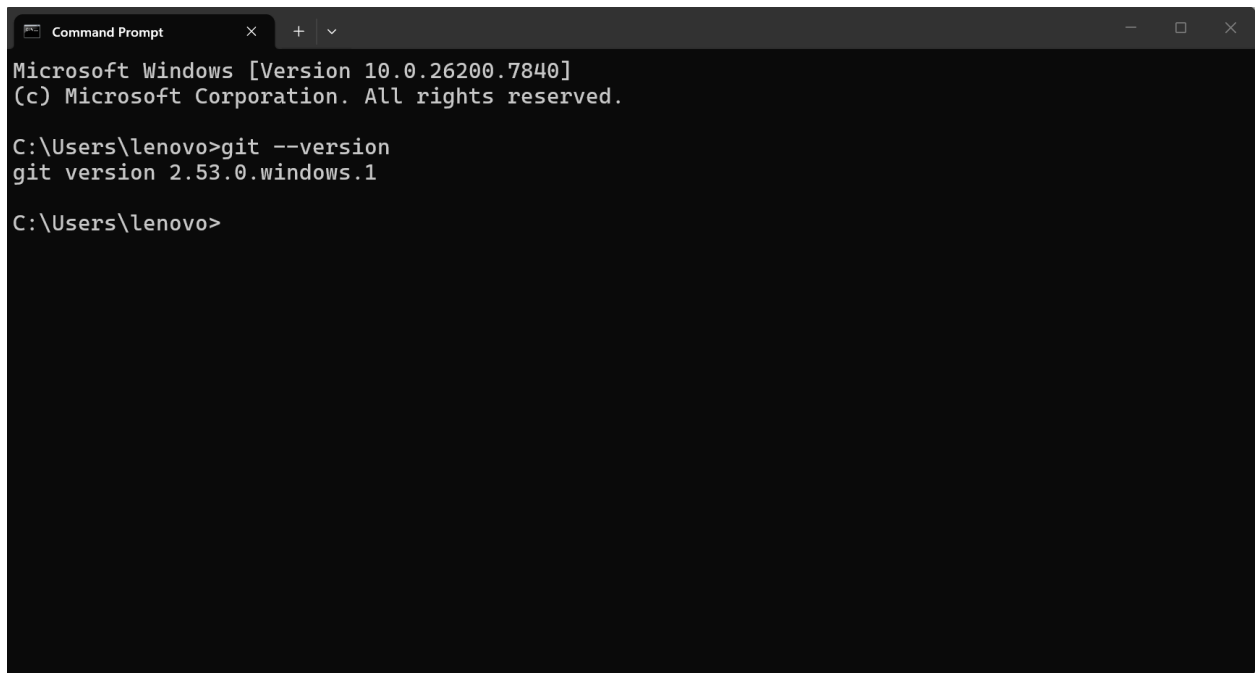


PART 2: Verify Git Installation

Step 14: Check Git Version

1. Open Command Prompt
2. Type : git - -version.

If installed correctly, you will see: git version 2.xx.x



```
Command Prompt
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>git --version
git version 2.53.0.windows.1

C:\Users\lenovo>
```

Suppose if you face error :

✗ Error: 'git' is not recognized

✓ Solution:

- Restart system, Reinstall Git, Make sure PATH option was selected correctly

PART 3: First-Time Git Configuration (Very Important)

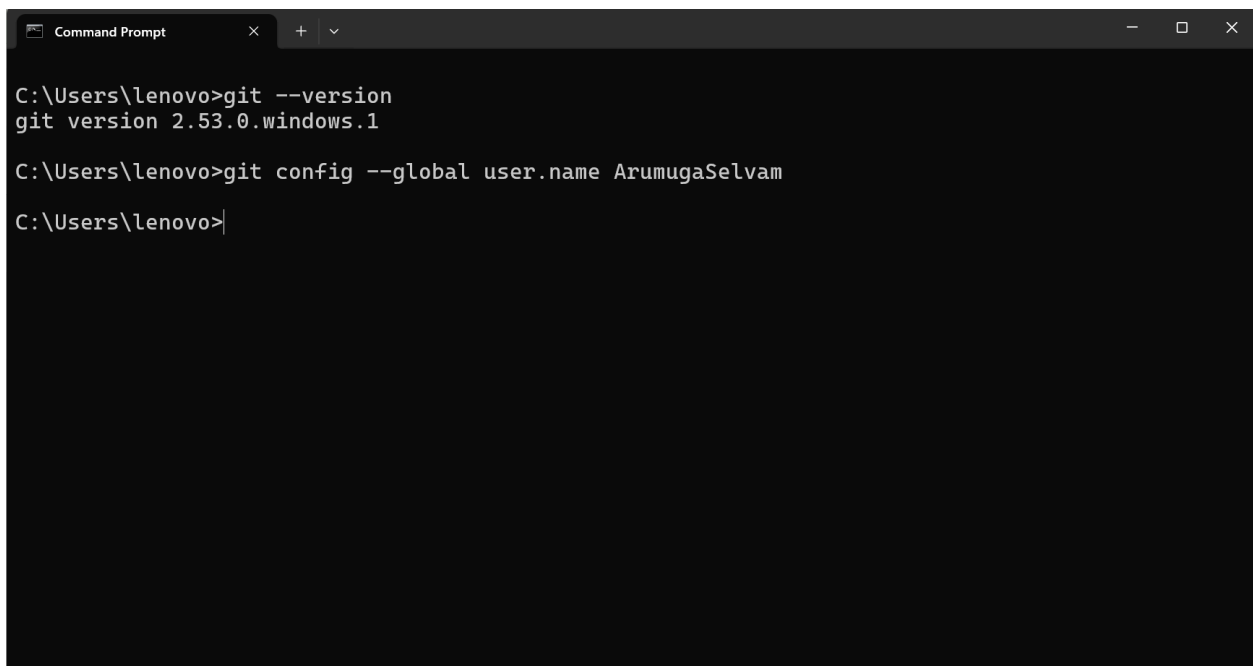
Step 15: Set Your Name

1. Open Command Prompt and type:

git config --global user.name "Your Full Name"

Example :

git config --global user.name "Arumugaselvam"

A screenshot of a Windows Command Prompt window. The title bar shows 'Command Prompt' with standard window controls. The command history is as follows:
C:\Users\lenovo>git --version
git version 2.53.0.windows.1

C:\Users\lenovo>git config --global user.name ArumugaSelvam

C:\Users\lenovo>|

```
C:\Users\lenovo>git --version
git version 2.53.0.windows.1

C:\Users\lenovo>git config --global user.name ArumugaSelvam

C:\Users\lenovo>|
```

Step 16: Set Your Email

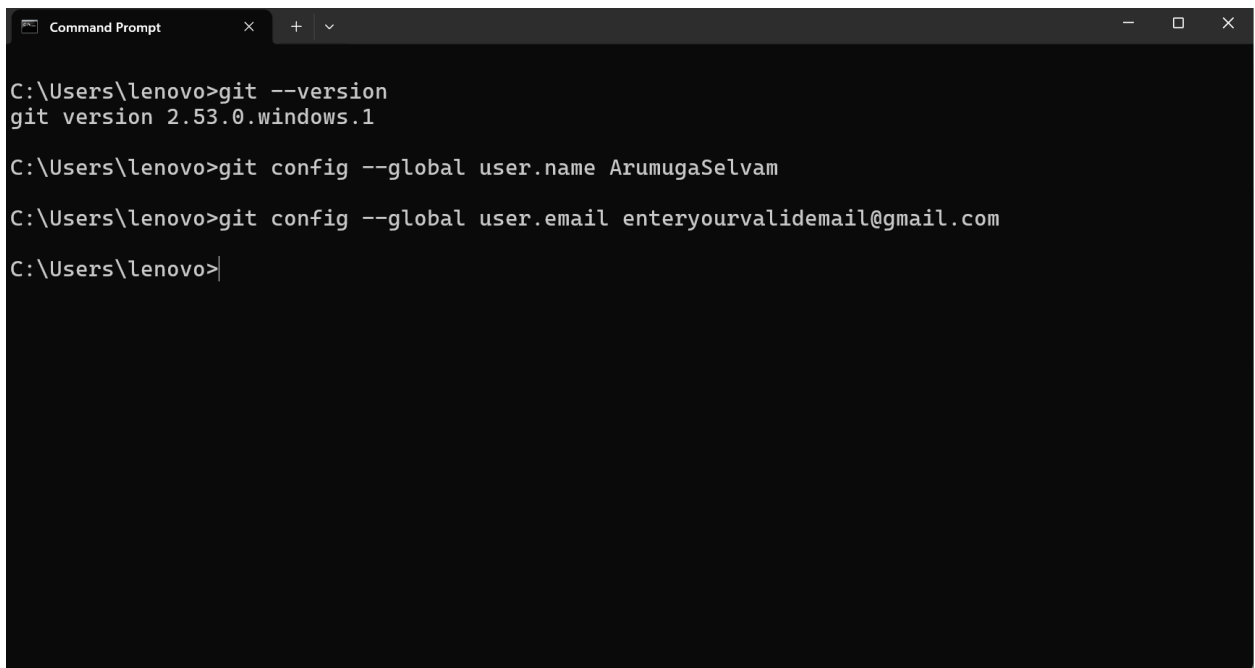
2. Open Command Prompt and type:

```
git config --global user.name "Your email@gmail.com"
```

 **Use the same email that you will use for GitHub.**

Example :

```
git config --global user.name "myemail@gmail.com"
```



```
Command Prompt
C:\Users\lenovo>git --version
git version 2.53.0.windows.1

C:\Users\lenovo>git config --global user.name ArumugaSelvam

C:\Users\lenovo>git config --global user.email enteryourvalidemail@gmail.com

C:\Users\lenovo>
```

Step 17: Verify Configuration

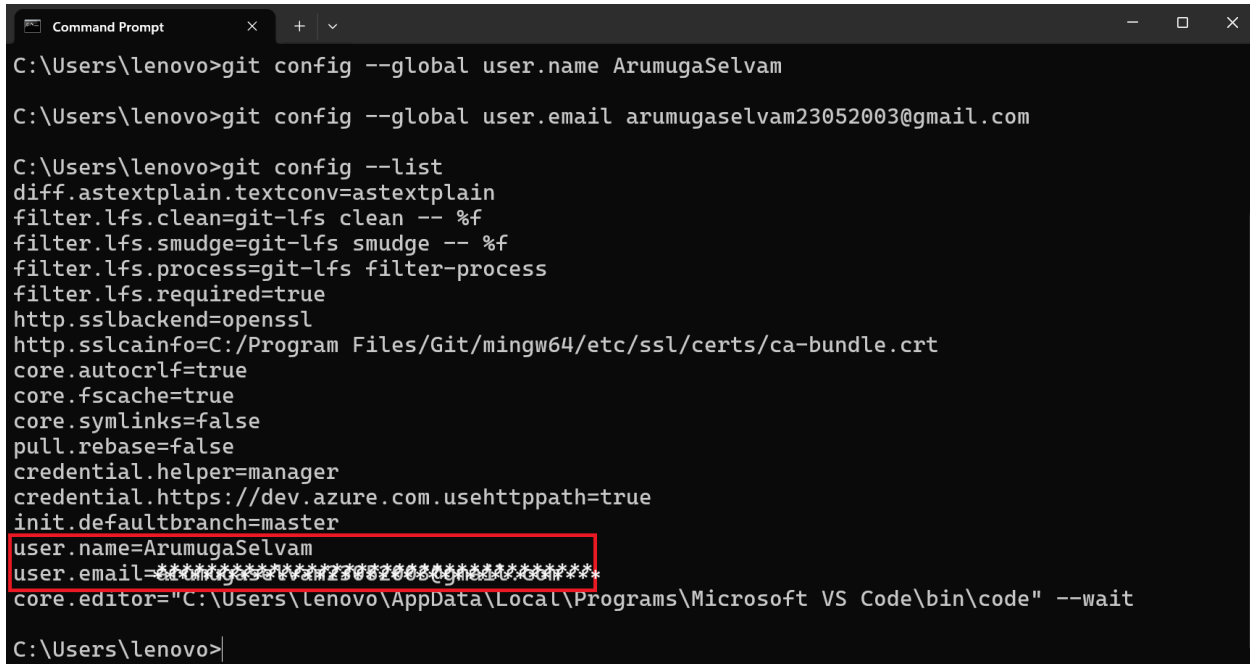
1. To check if name and email are set:

git config --list

You should see:

user.name=Your Name

user.email=your-email@gmail.com



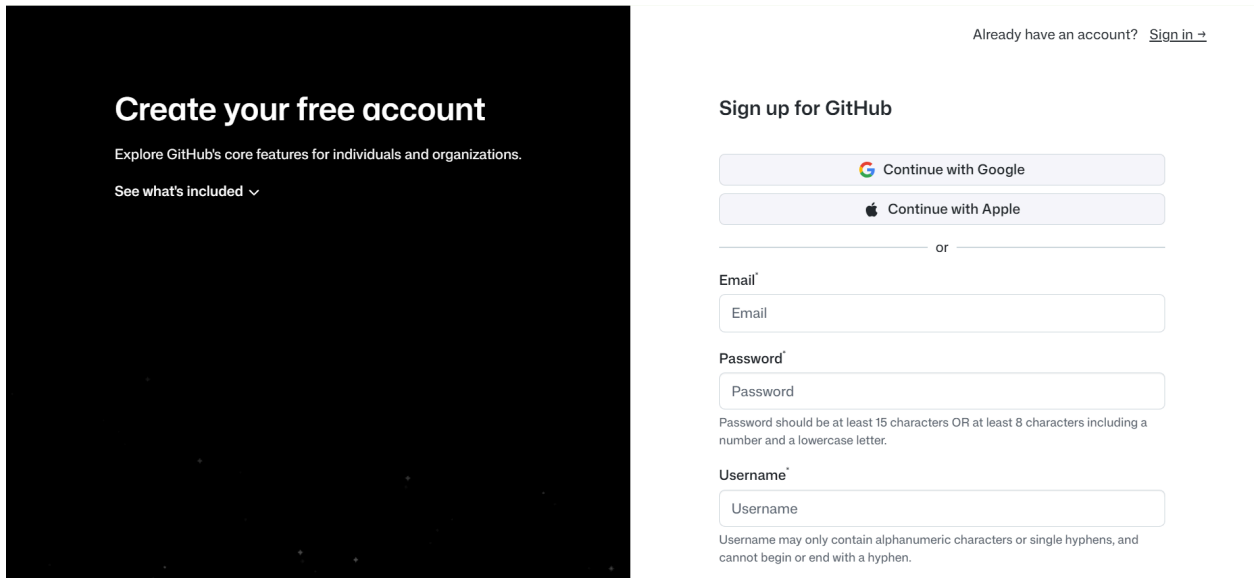
```
Command Prompt
C:\Users\lenovo>git config --global user.name ArumugaSelvam
C:\Users\lenovo>git config --global user.email arumugaselvam23052003@gmail.com
C:\Users\lenovo>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=ArumugaSelvam
user.email=arumugaselvam23052003@gmail.com
core.editor="C:\Users\lenovo\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
C:\Users\lenovo>
```

2. GitHub Account Creation

PART 1: Creating GitHub Account

Step 1: Open GitHub Website

1. Open your browser
2. Go to: <https://github.com>
3. Click on **Sign up**



The screenshot shows the GitHub 'Create your free account' page. On the left, a dark sidebar contains the heading 'Create your free account', a subtext 'Explore GitHub's core features for individuals and organizations.', and a link 'See what's included' with a dropdown arrow. The main content area is white and titled 'Sign up for GitHub'. It features two large buttons: 'Continue with Google' and 'Continue with Apple'. Below these is a link 'or' followed by three input fields: 'Email', 'Password', and 'Username'. The 'Email' field has a placeholder 'Email'. The 'Password' field has a placeholder 'Password' and a note below it: 'Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.' The 'Username' field has a placeholder 'Username' and a note below it: 'Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.' At the top right of the main area, there is a link 'Already have an account? Sign in' with an arrow.

Step 2: Enter Email Address

- Enter your email address
- Click **Continue**



Important:

Use a valid email address.

This email will be used for Git configuration later.

Step 3: Create Password

- Create a strong password
- Click **Continue**

Step 4: Choose Username

- Enter a unique username
- Click **Continue**



Important:

Username cannot be changed easily later.

Choose something professional.

Step 5: Email Preferences

Select your preference (can choose No).

Click **Continue**

Step 6: Verify Account

GitHub may ask:

- Puzzle verification
- Email OTP verification

Go to your email inbox and verify your account.

PART 2: Choose Plan

Step 7: Select Plan

You will see plan options.

👉 Select:

Free Plan

Click **Continue**

PART 3: First Login

After account creation, you will be redirected to the dashboard.

PART 4: Important First-Time Settings

Click profile picture (top right)

Click **Settings**

Add:

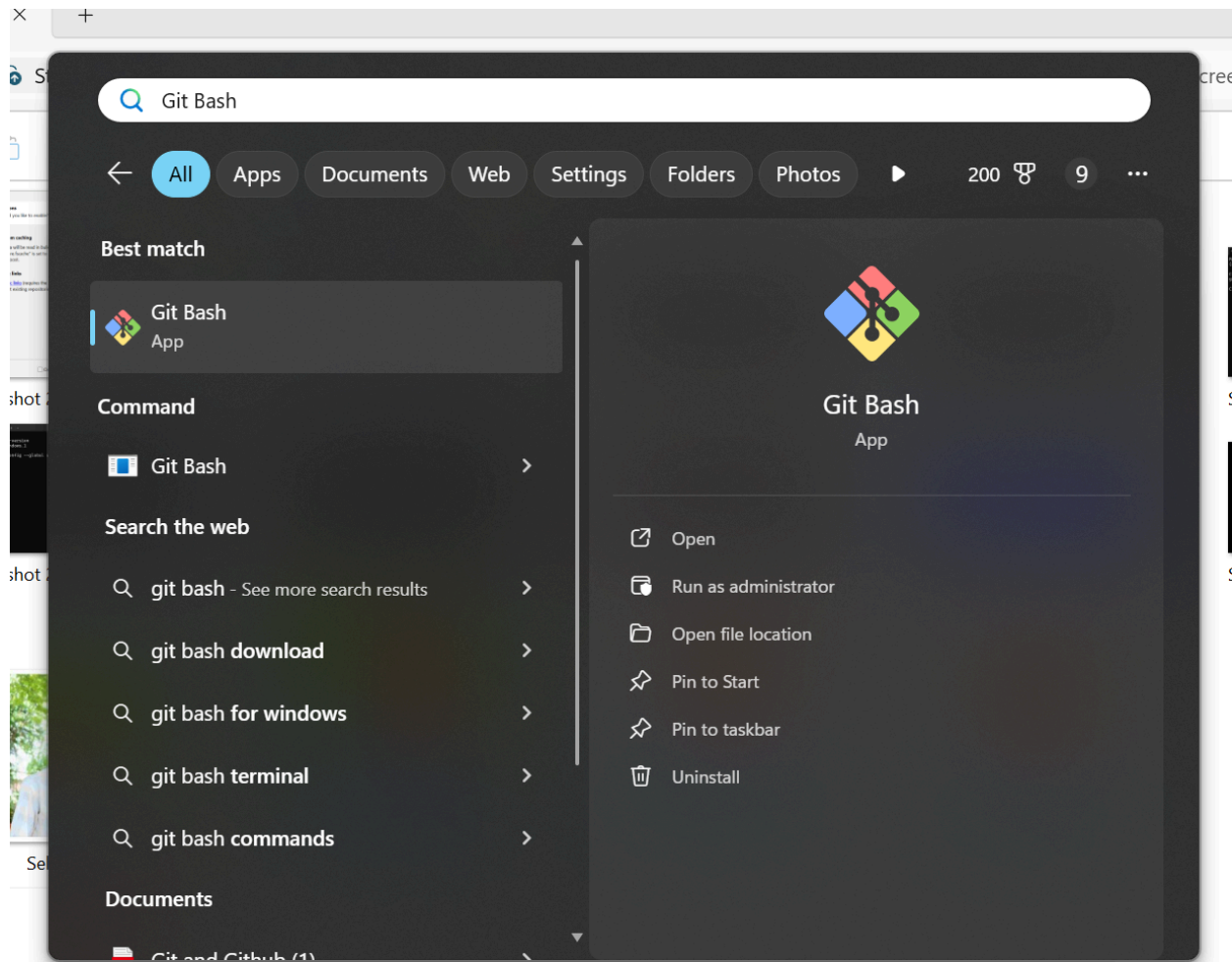
- Full Name
- Bio
- Location (optional)
- Profile photo

3. Connecting Git to GitHub Using SSH (One-Time Setup)

PART 1: Generate SSH Key

Step 1: Open Git Bash

1. Search: **Git Bash** in your system.
2. Open it



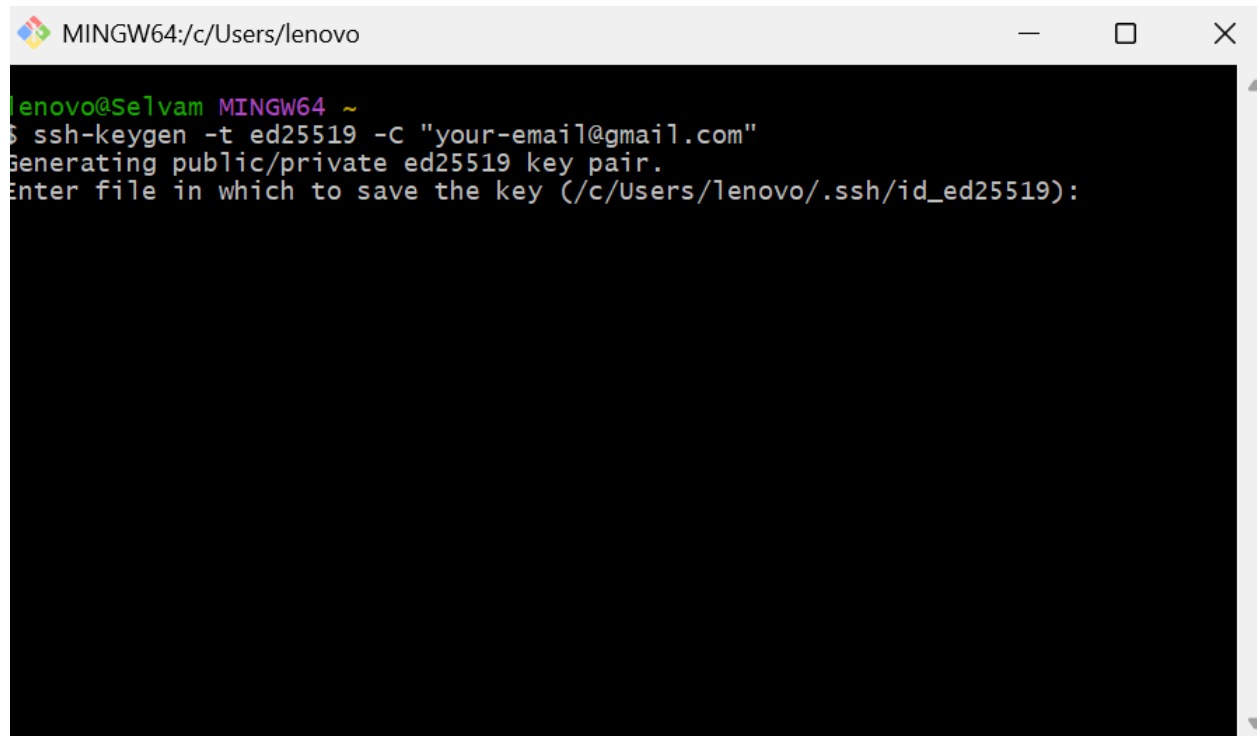
Step 2: Generate SSH Key

In Git Bash, type:

```
ssh-keygen -t ed25519 -C "your-email@gmail.com"
```

⚠ Use the same email you used for GitHub.

Press Enter when asked: Enter file in which to save the key

A screenshot of a Windows terminal window titled "MINGW64:/c/Users/lenovo". The terminal shows the command `ssh-keygen -t ed25519 -C "your-email@gmail.com"` being executed. The output indicates that a public/private ed25519 key pair is being generated. It then prompts the user to "Enter file in which to save the key (/c/Users/lenovo/.ssh/id_ed25519):". The terminal window has standard Windows window controls (minimize, maximize, close) in the top right corner.

```
lenovo@Selvam MINGW64 ~  
$ ssh-keygen -t ed25519 -C "your-email@gmail.com"  
Generating public/private ed25519 key pair.  
Enter file in which to save the key (/c/Users/lenovo/.ssh/id_ed25519):
```

Press Enter again for passphrase :

```
overwrite (y/n): y  
Enter passphrase for "/c/Users/lenovo/.ssh/id_ed25519" (empty for no passphrase)  
: |
```

Press Enter again for passphrase again :

```
overwrite (y/n): y  
Enter passphrase for "/c/Users/lenovo/.ssh/id_ed25519" (empty for no passphrase)  
:  
Enter same passphrase again:
```

Once the SSH Key Generated. It will show output like this :

```
Enter same passphrase again:
Your identification has been saved in /c/Users/lenovo/.ssh/id_ed25519
Your public key has been saved in /c/Users/lenovo/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:EWqBFNVz5OnzP1CShnBV07cSiixXup/RL0LpBfzzhyM your-email@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|    .oo..o.o.    |
|   oo.o ...o .   |
|  .o=oo* . o .   |
|  .o=Oo= . .     |
|   +S+ = .       |
|   .o + =        |
|   o= + + .      |
|   o* E = .      |
|   .oo o o       |
+-----[SHA256]-----+

lenovo@Selvam MINGW64 ~
$
```

Step 3: Locate the SSH Key

Run :

```
cat ~/.ssh/id_ed25519.pub
```

You will see a long key starting with:

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI....
```

👉 Copy the entire key.

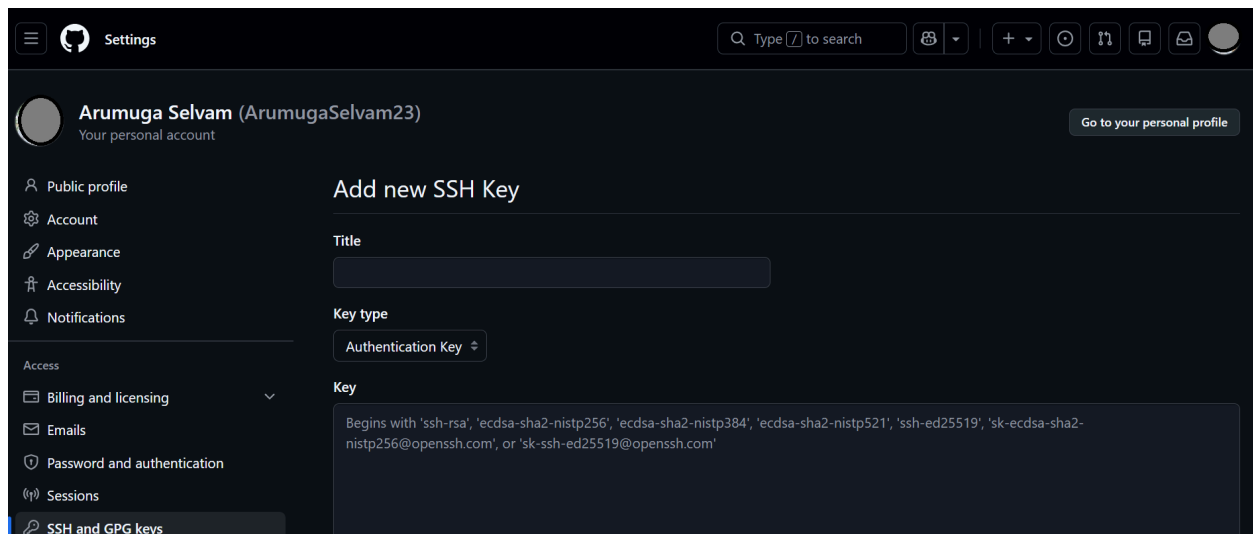
```
lenovo@Selvam MINGW64 ~
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIO6Eot4Om/Oxam6/emA7yv1808nwQarQbQac7kewzkjv
your-email@gmail.com

lenovo@Selvam MINGW64 ~
$
```

PART 2: Add SSH Key to GitHub

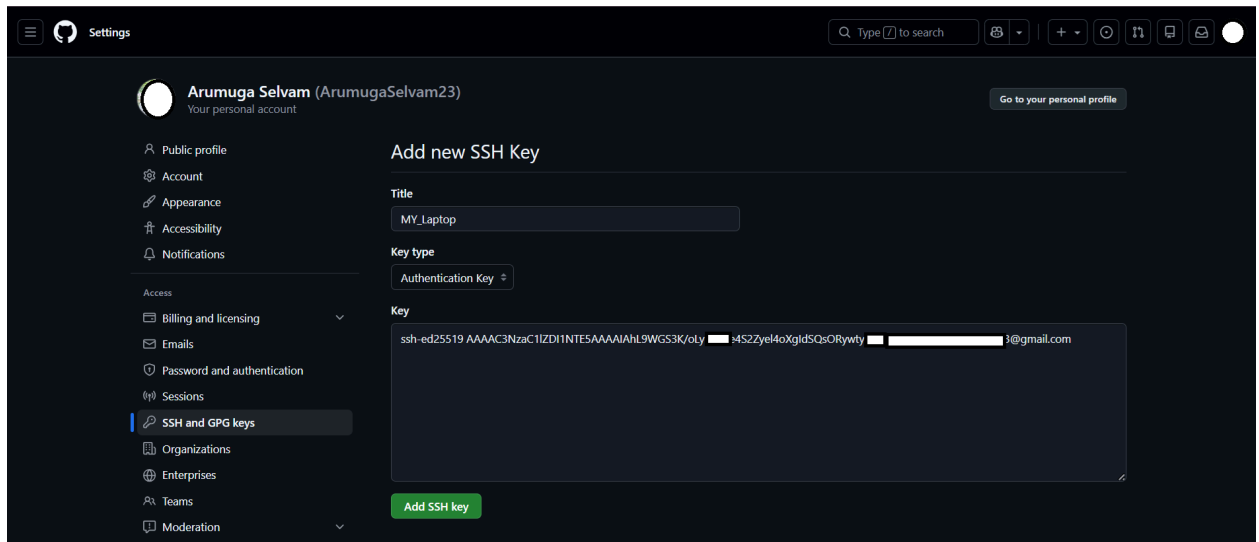
Step 4: Open GitHub Settings

1. Go to <https://github.com>
2. Click profile picture (top right)
3. Click **Settings**
4. Click **SSH and GPG keys** (on left side)
5. Click **New SSH key** (on right side)



Step 5: Add the Key

1. Title: My Laptop
2. Key type: Authentication Key
3. Paste the copied SSH key
4. Click Add SSH key



Step 6: Test the Connection

In Git Bash, type:

```
ssh -T git@github.com
```

You may see: Are you sure you want to continue connecting?

Type : yes

Press Enter.

If successful, you will see:

Hi username! You've successfully authenticated...

```
lenovo@Selvam MINGW64 ~  
$ ssh -T git@github.com  
Hi ArumugaSelvam23! You've successfully authenticated, but GitHub does not provide shell access.  
  
lenovo@Selvam MINGW64 ~  
$ |
```

Important Notes

- ✓ This setup is done only once.
- ✓ Do not share private key with anyone.
- ✓ SSH is like creating a secure identity card for your computer. Once GitHub recognizes your computer, it won't ask password again."

4.Creating First Repository and Pushing Code to GitHub

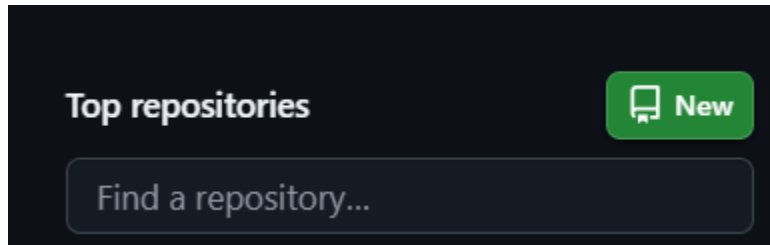
PART 1: Create a Repository on GitHub

Step 1: Open GitHub

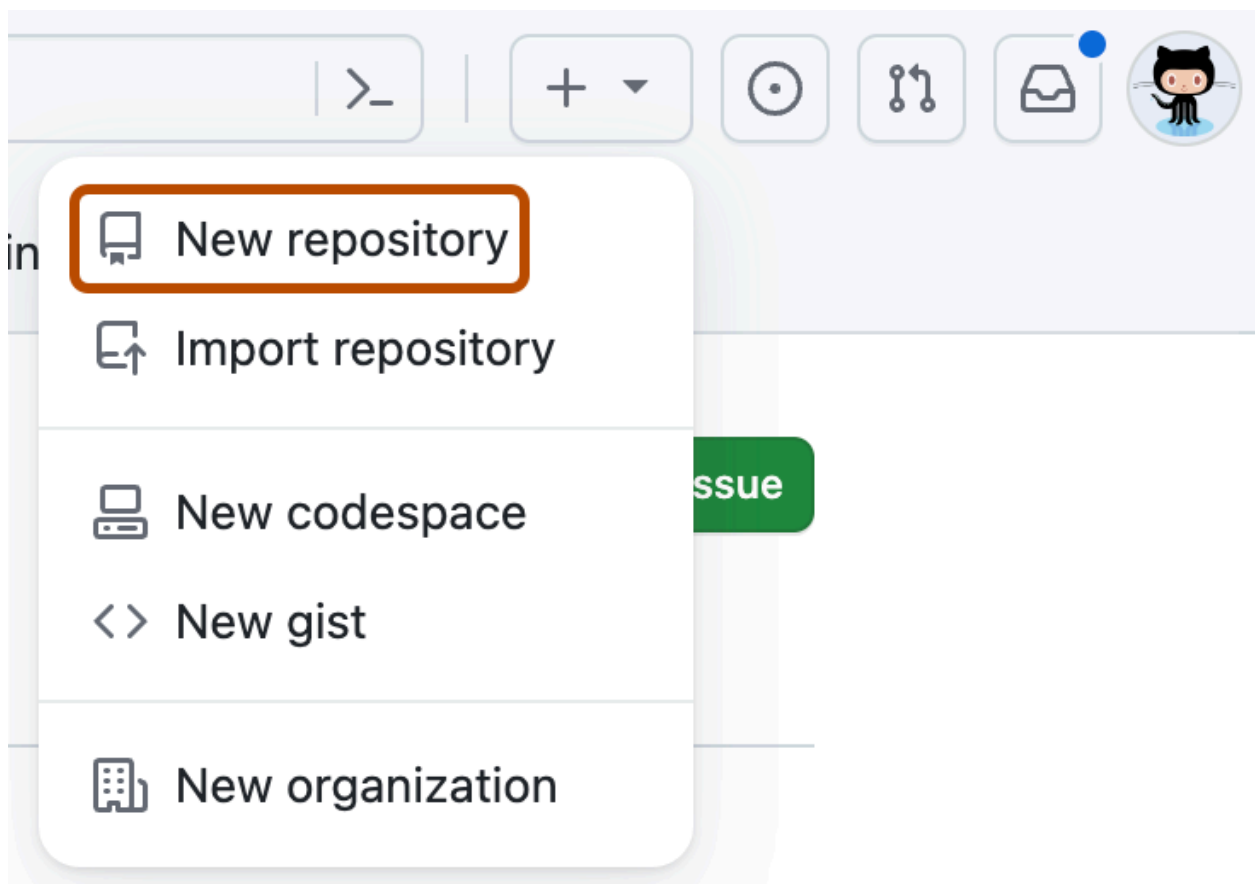
1. Go to <https://github.com>
2. Login to your account

Step 2: Create New Repository

(Click New or New repository option)



(OR)



Step 3: Fill Repository Details

Repository Name: first-project

Description : type about your repo.

Visibility: Public

⚠ Do NOT check **Add README** (important for beginners).

Then click: 🖱 **Create Repository**

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

ArumugaSelvam23

Repository name *

wsa-internship-project

✔ wsa-internship-project is available.

Great repository names are short and memorable. How about [sturdy-spork](#)?

Description

Repo for WSA Internship Project

31 / 350 characters

2

Configuration

Choose visibility *

Public

Choose who can see and commit to this repository

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

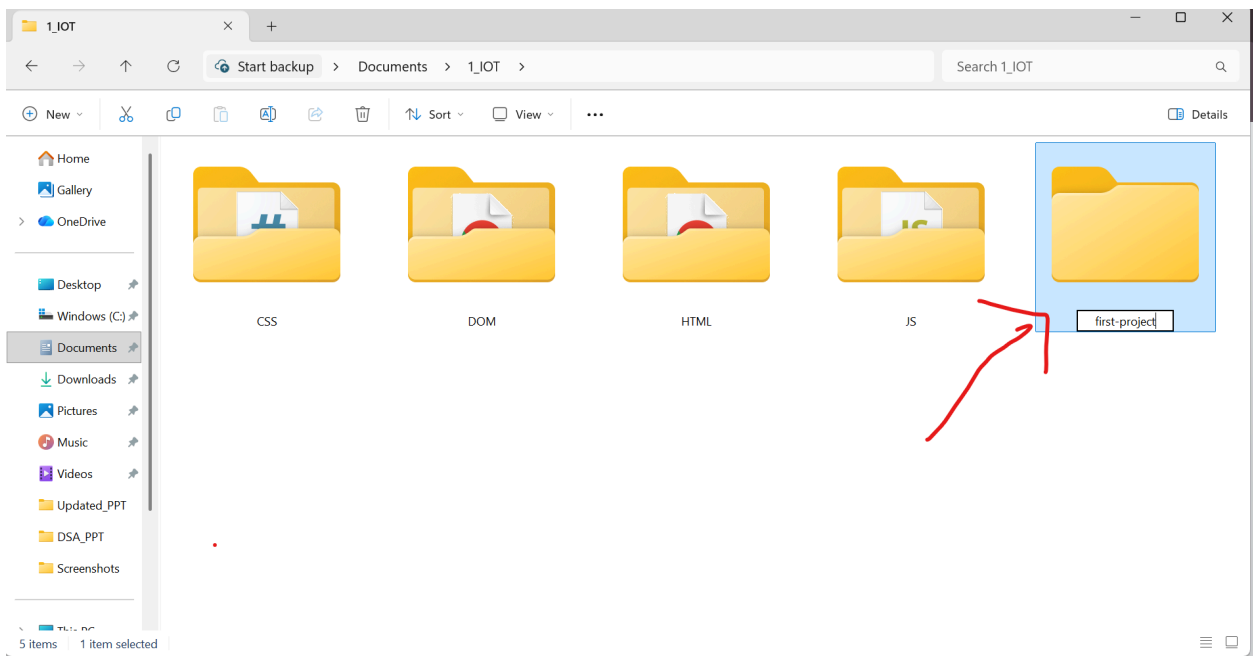
Create repository

PART 2: Create Local Project

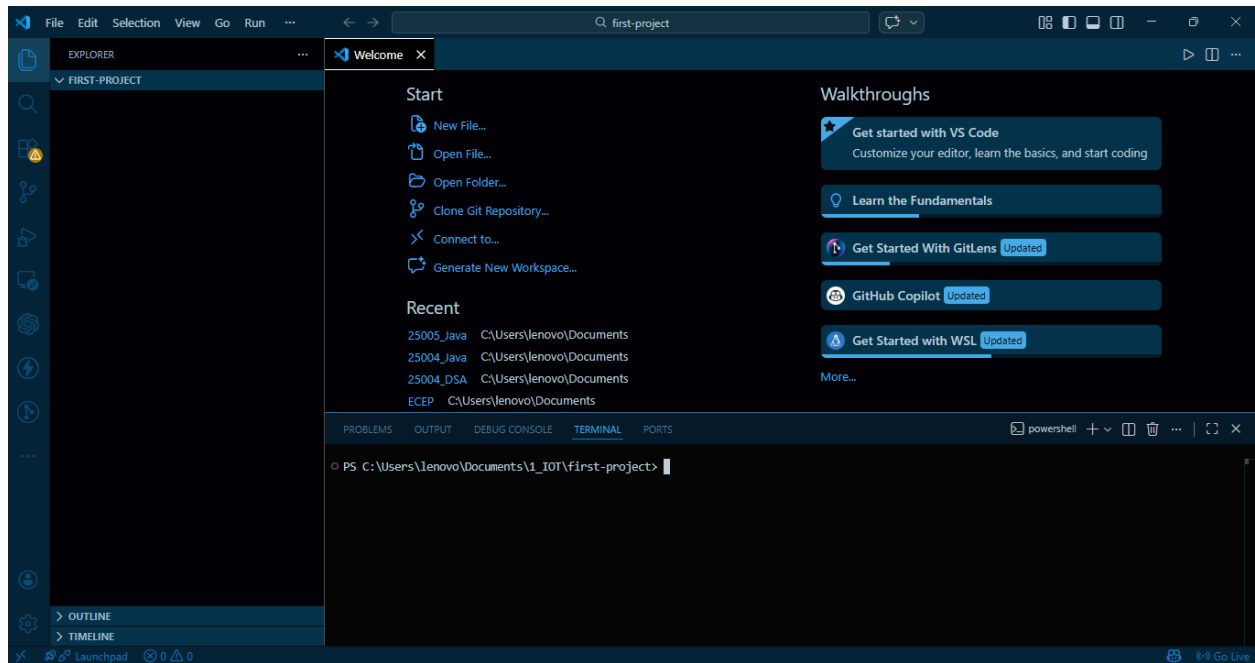
Step 4: Create Project Folder

On your computer:

1. Create a new folder
Example: first-project

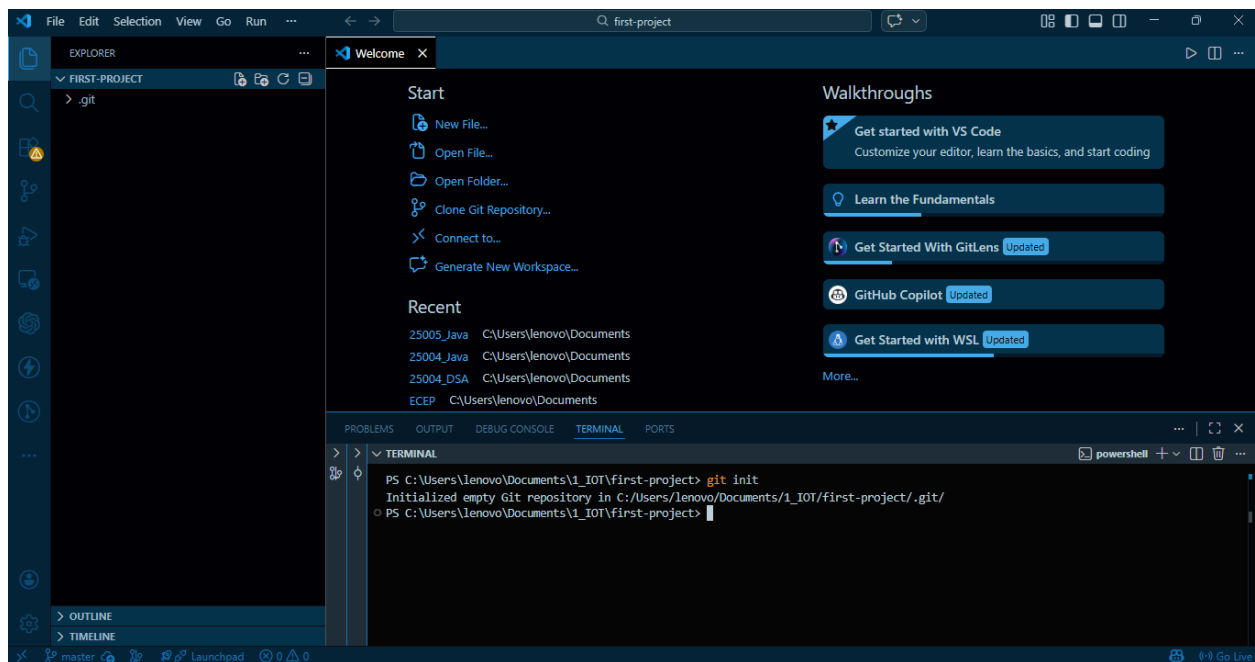


Step 5: Open this folder in your VS Code and also open the terminal.



Step 6: Initialize Git

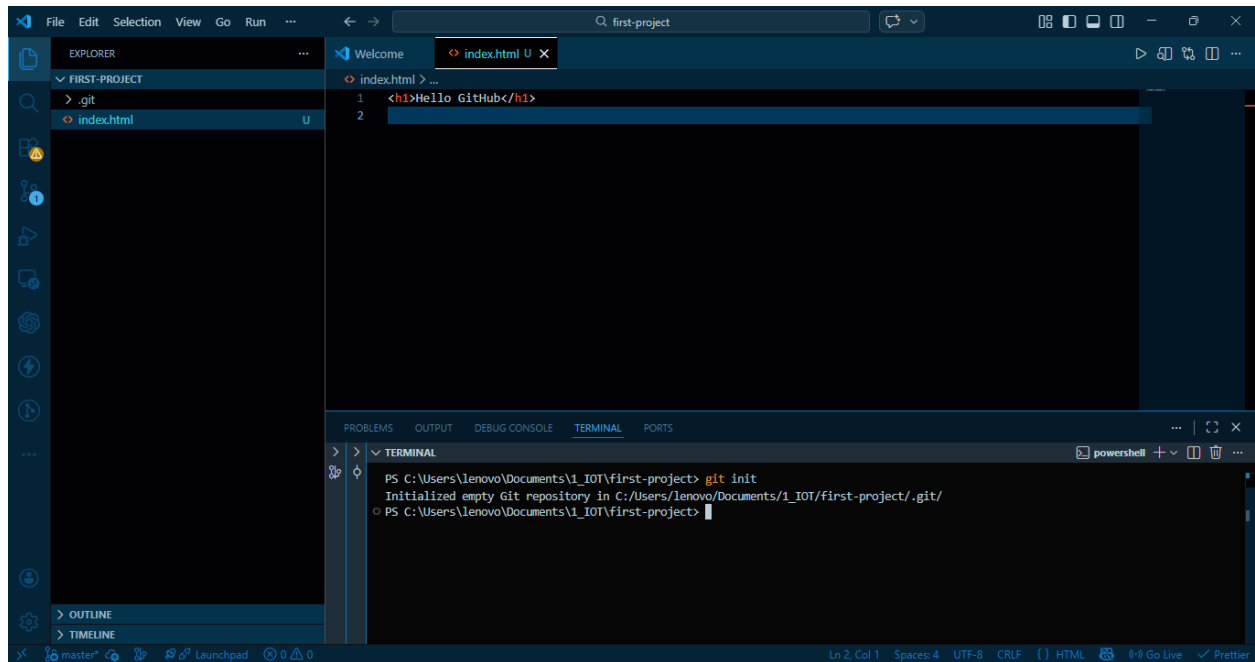
Run : `git init`



Step 7: Create a Sample File

Create a file: **index.html**

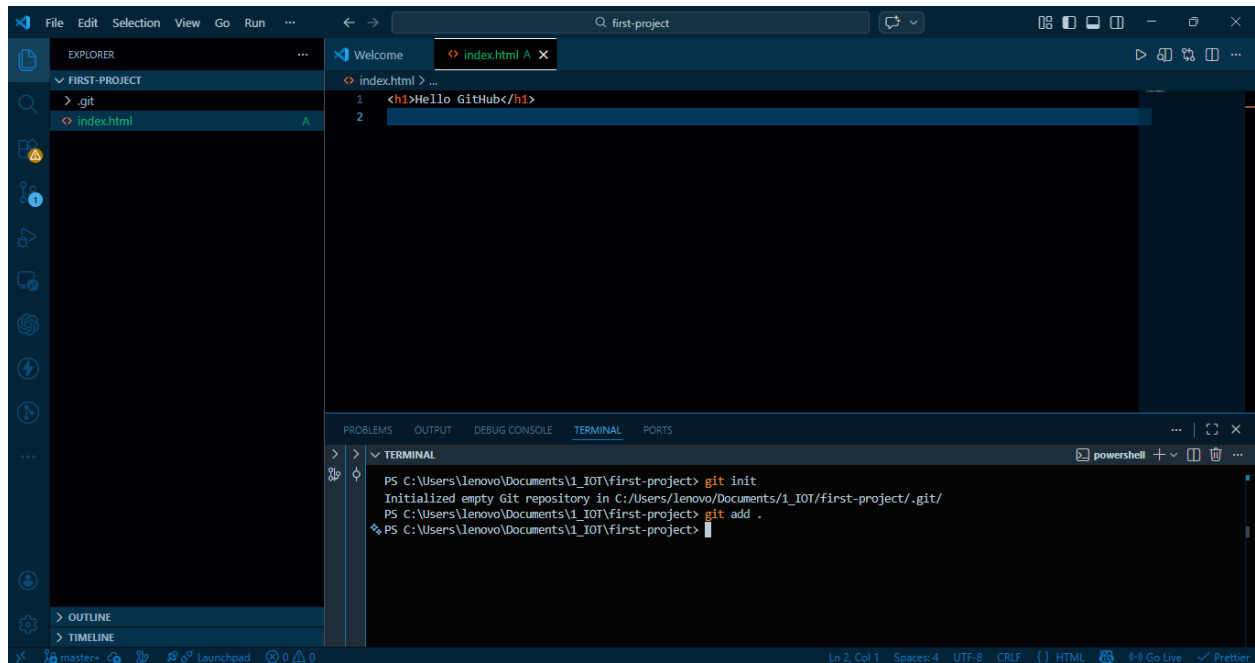
Example content: **<h1>Hello GitHub</h1>**



Step 8: Add Files to Git

Run : **git add .**

This stages all files.



The screenshot shows the Visual Studio Code interface with a project named 'first-project'. The Explorer panel on the left shows a file structure with a '.git' directory and an 'index.html' file. The editor window displays the content of 'index.html', which is a simple HTML document with a single line: `<h1>Hello GitHub</h1>`. The terminal panel at the bottom shows the execution of the following commands:

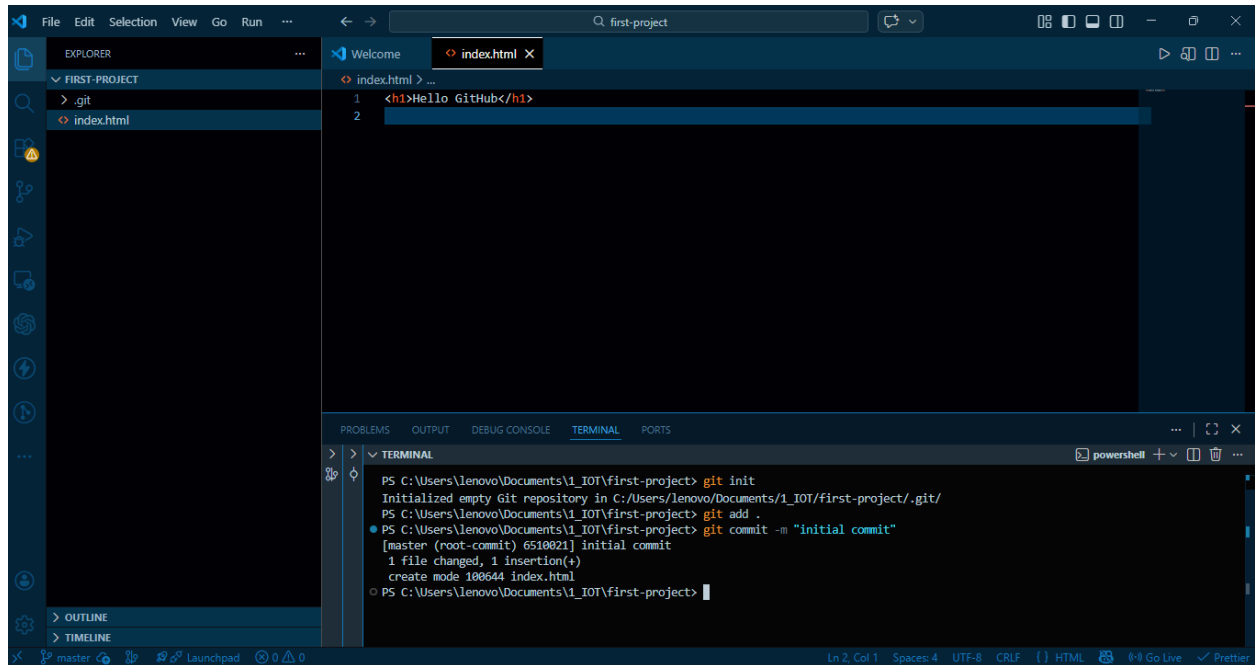
```
PS C:\Users\lenovo\Documents\1_IOT\first-project> git init
Initialized empty Git repository in C:/Users/lenovo/Documents/1_IOT/first-project/.git/
PS C:\Users\lenovo\Documents\1_IOT\first-project> git add .
PS C:\Users\lenovo\Documents\1_IOT\first-project>
```

The status bar at the bottom indicates the current file is 'index.html' and the editor is using the 'Prettier' formatter.

Step 9: Commit Changes

Run: **git commit -m "Initial commit"**

This saves your changes in Git history.

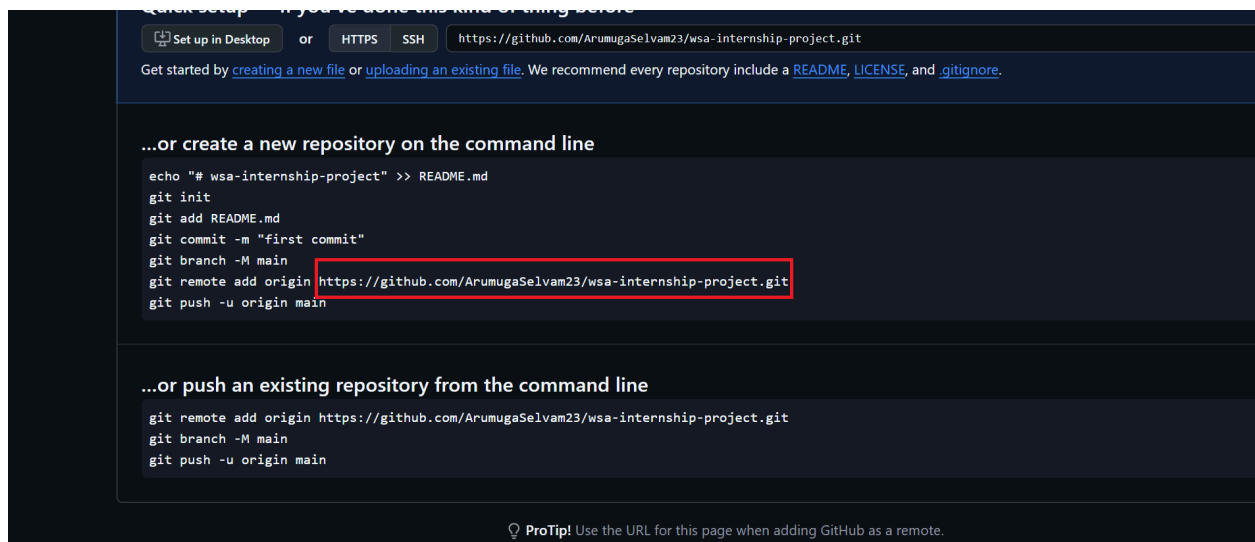
A screenshot of the Visual Studio Code editor interface. The Explorer pane on the left shows a project named 'FIRST-PROJECT' with files '.git' and 'index.html'. The editor pane shows 'index.html' with the content:

```
1 <h1>Hello GitHub</h1>
2
```

 The TERMINAL pane at the bottom shows the following commands and output:

```
PS C:\Users\lenovo\Documents\1_IOT\first-project> git init
Initialized empty Git repository in C:/Users/lenovo/Documents/1_IOT/first-project/.git/
PS C:\Users\lenovo\Documents\1_IOT\first-project> git add .
PS C:\Users\lenovo\Documents\1_IOT\first-project> git commit -m "initial commit"
[master (root-commit) 6510021] initial commit
1 file changed, 1 insertion(+)
 create mode 100644 index.html
PS C:\Users\lenovo\Documents\1_IOT\first-project>
```

Step 10: Copy Repository URL from Github

A screenshot of the GitHub website's 'Quickstart' page. It shows instructions for creating a new repository. The URL `https://github.com/ArumugaSelvam23/wsa-internship-project.git` is highlighted with a red box. The commands shown are:

```
...or create a new repository on the command line

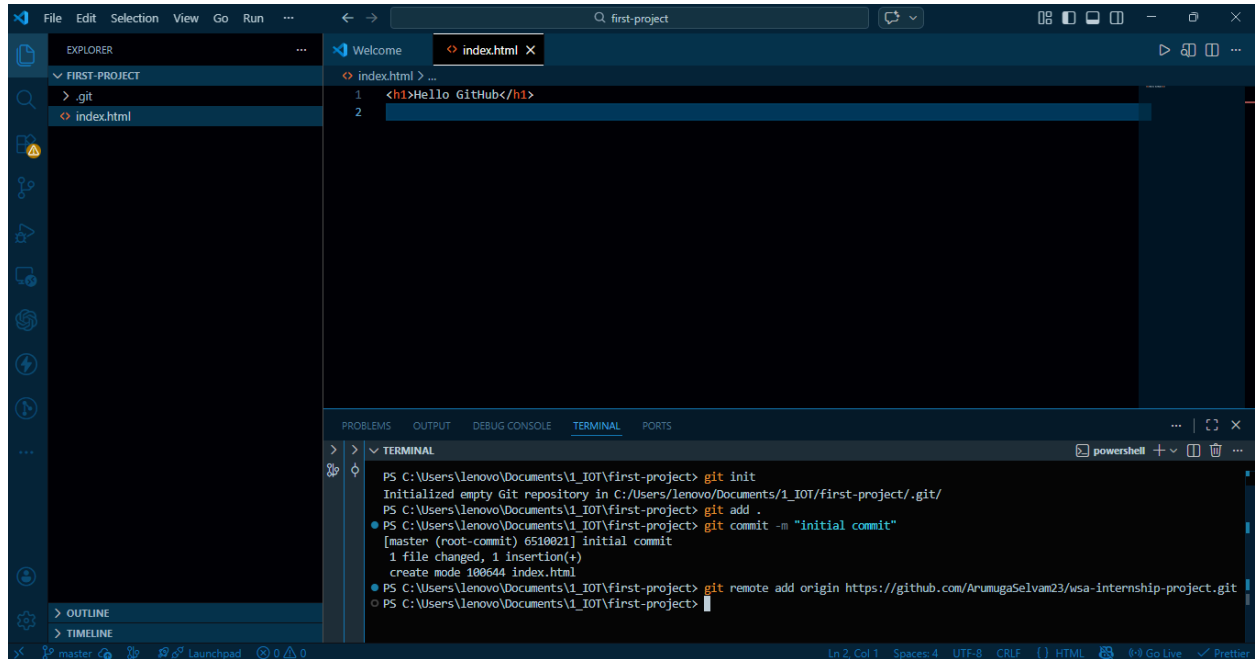
echo "# wsa-internship-project" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/ArumugaSelvam23/wsa-internship-project.git
git push -u origin main

...or push an existing repository from the command line

git remote add origin https://github.com/ArumugaSelvam23/wsa-internship-project.git
git branch -M main
git push -u origin main
```

Step 11: Add Remote Repository

Run : **git remote add origin “pasteyourcopiedurl”**



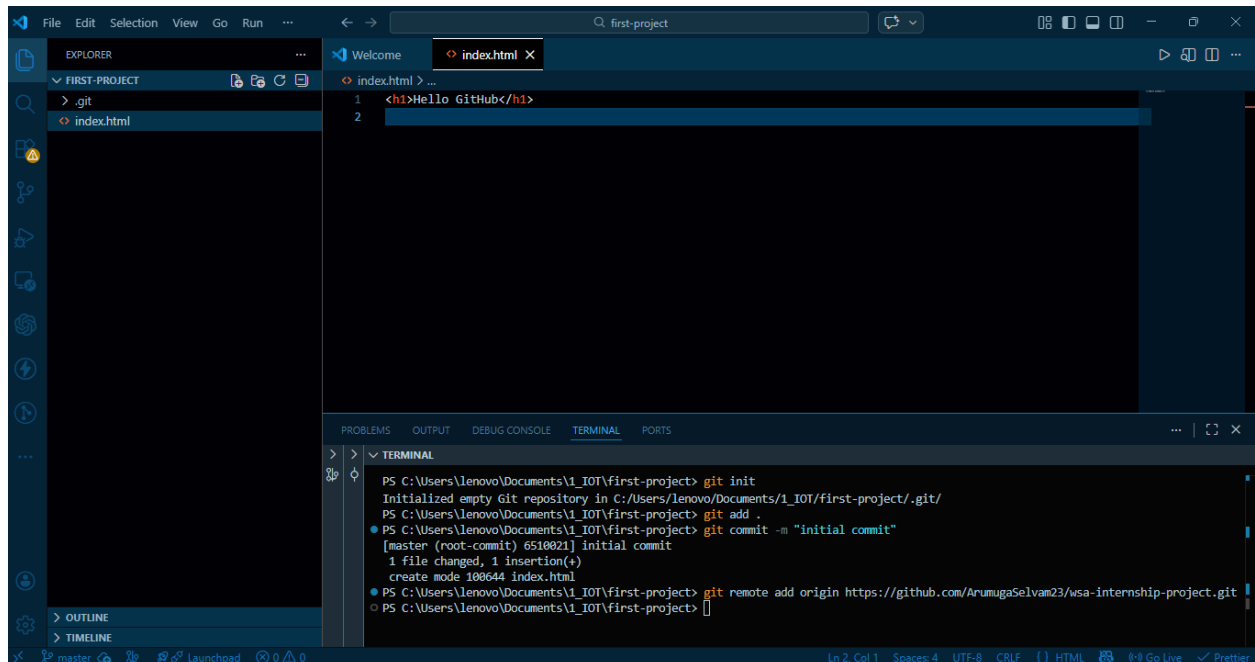
The screenshot shows the Visual Studio Code interface with a file explorer on the left showing a project named 'FIRST-PROJECT' containing a '.git' directory and an 'index.html' file. The main editor displays 'index.html' with the content:

```
1 <h1>Hello GitHub</h1>
2
```

 The bottom terminal panel shows the following commands and output:

```
PS C:\Users\lenovo\Documents\1_IOT\first-project> git init
Initialized empty Git repository in C:/Users/lenovo/Documents/1_IOT/first-project/.git/
PS C:\Users\lenovo\Documents\1_IOT\first-project> git add .
PS C:\Users\lenovo\Documents\1_IOT\first-project> git commit -m "initial commit"
[master (root-commit) 6518021] initial commit
1 file changed, 1 insertion(+)
create mode 100644 index.html
PS C:\Users\lenovo\Documents\1_IOT\first-project> git remote add origin https://github.com/ArumugaSelvam23/wsa-internship-project.git
PS C:\Users\lenovo\Documents\1_IOT\first-project>
```

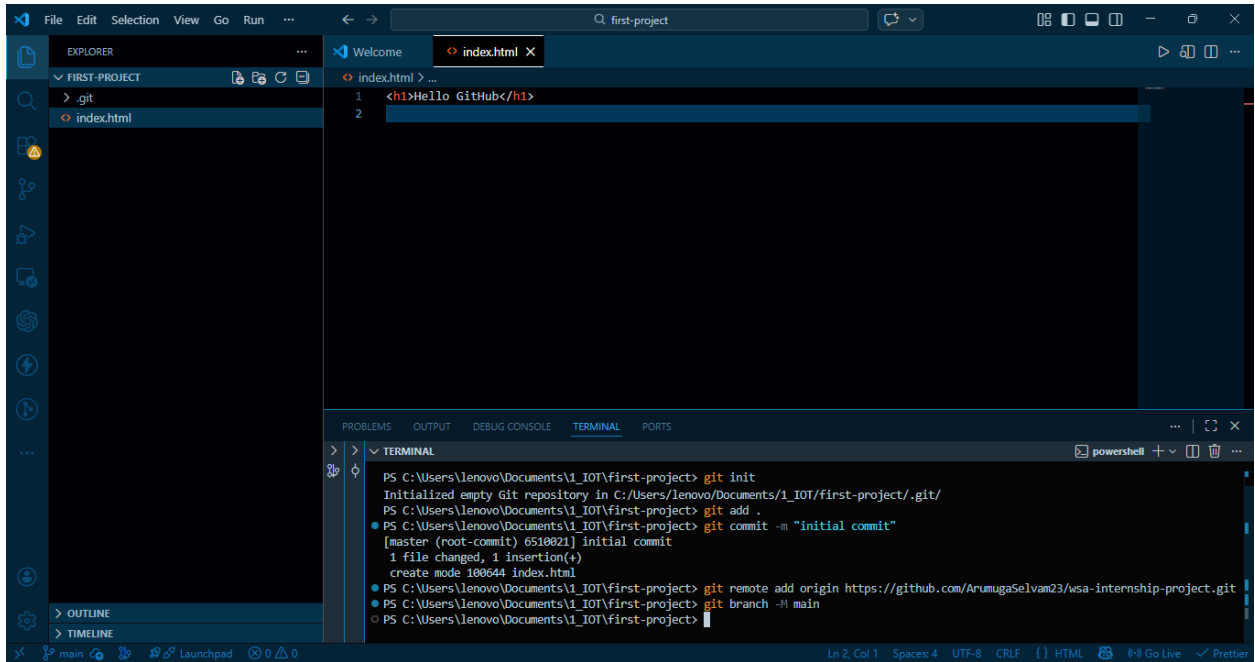
Run : **git remote add origin “pasteyourcopiedurl”**



This screenshot is identical to the one above, showing the same VS Code interface and terminal output for Step 11. The terminal shows the execution of 'git init', 'git add .', 'git commit -m "initial commit"', and 'git remote add origin https://github.com/ArumugaSelvam23/wsa-internship-project.git'.

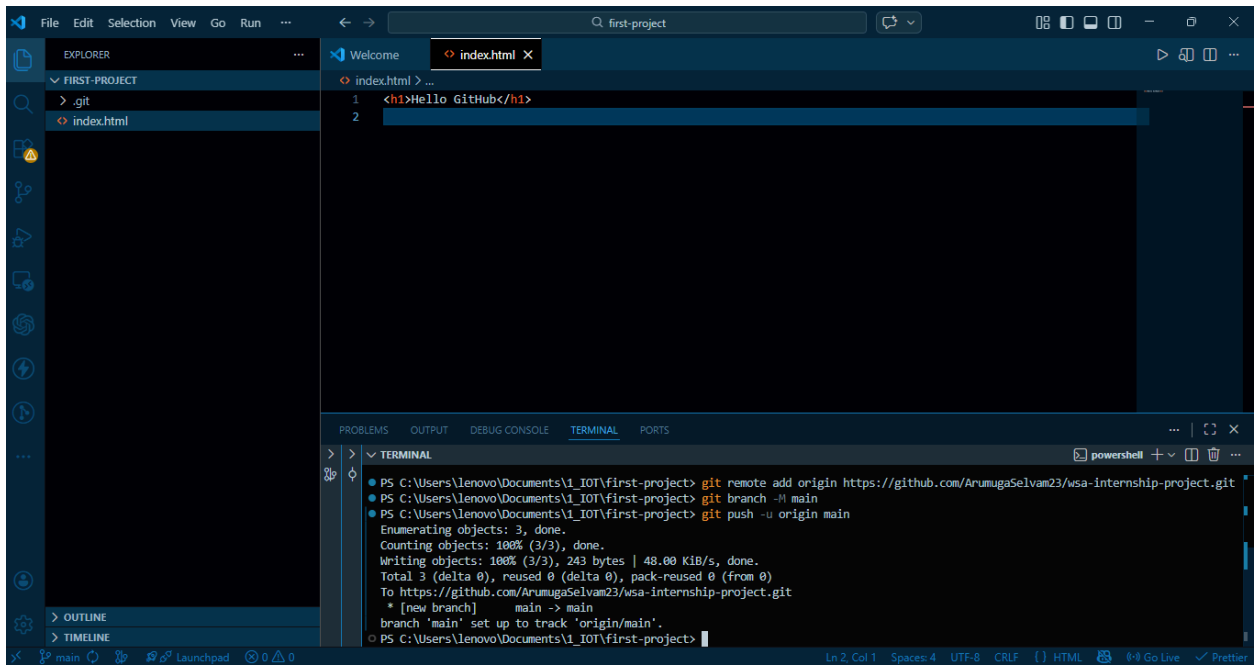
Step 12: Push Code to GitHub

Run : **git branch -M main**



```
PS C:\Users\lenovo\Documents\1_IOT\first-project> git init
Initialized empty Git repository in C:/Users/lenovo/Documents/1_IOT/first-project/.git/
PS C:\Users\lenovo\Documents\1_IOT\first-project> git add .
PS C:\Users\lenovo\Documents\1_IOT\first-project> git commit -m "initial commit"
[master (root-commit) 6510021] initial commit
1 file changed, 1 insertion(+)
 create mode 100644 index.html
PS C:\Users\lenovo\Documents\1_IOT\first-project> git remote add origin https://github.com/ArumugaSelvam23/wsa-internship-project.git
PS C:\Users\lenovo\Documents\1_IOT\first-project> git branch -M main
```

Then: **git push -u origin main**



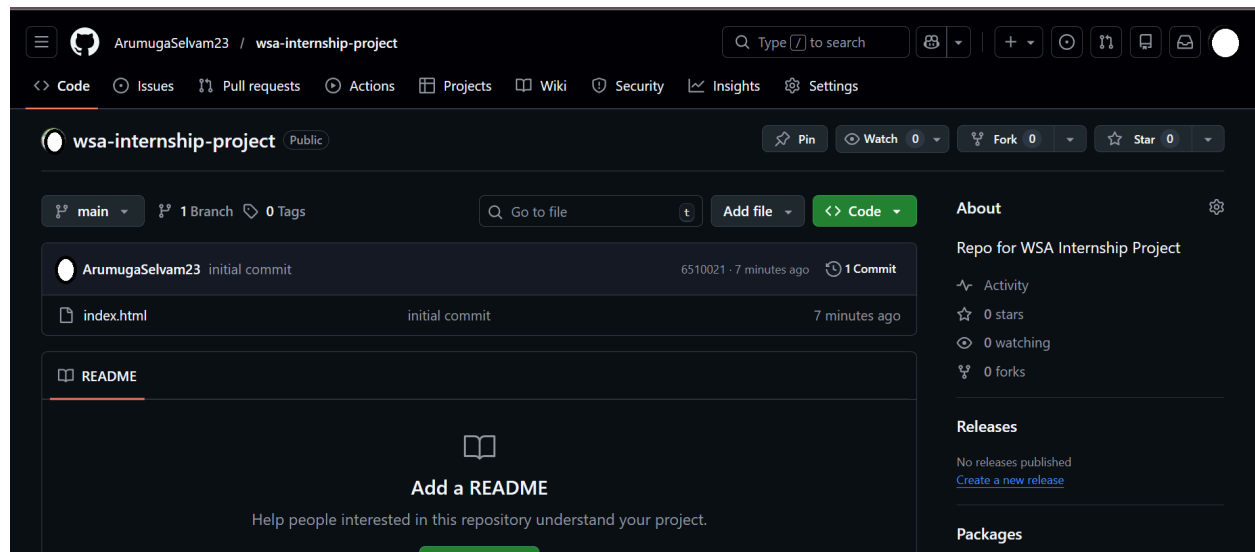
```
PS C:\Users\lenovo\Documents\1_IOT\first-project> git remote add origin https://github.com/ArumugaSelvam23/wsa-internship-project.git
PS C:\Users\lenovo\Documents\1_IOT\first-project> git branch -M main
PS C:\Users\lenovo\Documents\1_IOT\first-project> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 243 bytes | 48.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ArumugaSelvam23/wsa-internship-project.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\lenovo\Documents\1_IOT\first-project>
```



Success!

Go to your GitHub repository page and refresh.

You will see your files uploaded.



"Every project you build from today should be pushed to GitHub. It helps you to build a strong portfolio." 👍

⚠️ Note:

This installation guide is intended only for the first-time setup of Git and GitHub on your system. You do not need to repeat these steps every time.

For updating or submitting your project code, please follow the instructions provided by your instructor during the course.

If you would like to learn more about Git and GitHub, you may refer to the additional learning resources provided in LMS Section.